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Opposite and complementary roles of the two calcium thresholds for inducing LTP and LTD in models of striatal projection neurons

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This computational study constitutes an extension to prior work on biophysical calcium-based synaptic plasticity rules with metaplasticity, investigating how single neurons can learn to perform non-linear pattern classification. This **important** work presents a significantly simpler solution to the studied problem with potentially broad applicability, there is however **incomplete** evidence to support the core conclusions.

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Abstract

Synaptic plasticity has been shown to occur when calcium, flowing into the synapse due to incoming stimuli, surpasses a threshold level. This threshold level is modifiable through a process called metaplasticity. Some neurons, such as the striatal projection neurons, use different sources of calcium as the signal for synaptic strengthening (long-term potentiation, LTP) or weakening (long-term depression, LTD), resulting in them having two thresholds for inducing plasticity. In this study, we show opposite and complementary roles of metaplasticity in these two thresholds for inducing LTP and LTD on learning how to solve the linear and nonlinear feature binding problem (FBP and NFBP). In short, metaplasticity in one threshold (e.g. LTD) allows synaptic plasticity *of the opposite type* (e.g. LTP) to be properly expressed. This happens because metaplasticity in the LTD threshold protects strengthened synapses from weakening, thus allowing them to persistently increase during learning (and encode learned patterns). Similarly, metaplasticity in the LTP threshold prevents weakened synapses from strengthening, thus allowing them to persistently decrease. Metaplasticity in both thresholds is necessary when synapses are clustered and the neuron needs to rely on supralinear dendritic integration for learning.

Introduction

Learning is thought to be implemented by several processes in the nervous system, one of which is synaptic plasticity - the modification of synapses based on incoming stimuli to the neuron (Citri and Malenka, 2007 [↗](#); Kandel et al., 2014 [↗](#); Takeuchi et al., 2014 [↗](#); Abraham et al., 2019 [↗](#)). Most often, incoming stimuli cause calcium influx in the synapse, an important “second messenger” ion that triggers synaptic plasticity. Importantly, not all levels of calcium can elicit synaptic plasticity - threshold levels of calcium have been reported above which plasticity is triggered (Yang et al., 1999 [↗](#); Nishiyama et al., 2000 [↗](#); Cormier et al., 2001 [↗](#); Shindou et al., 2011 [↗](#)). These calcium thresholds for inducing plasticity are themselves modifiable (plastic) through a process called metaplasticity (Abraham, 2008 [↗](#)). Metaplasticity regulates the conditions under which plasticity occurs, i.e. under which synapses are modified.

In many neurons, such as hippocampal CA1 neurons, Purkinje neurons, cerebellar Golgi cells, and projection neurons from the core of the nucleus accumbens, the calcium signal for synaptic strengthening (long-term potentiation, LTP) and for synaptic weakening (long-term depression, LTD) comes from different sources (Nishiyama et al., 2000 [↗](#); Kohda et al., 1995 [↗](#); Binda et al., 2016 [↗](#); Locatelli et al., 2021 [↗](#); Ji and Martin, 2012 [↗](#); Cho et al., 2001 [↗](#)). In the direct-pathway striatal projection neurons (dSPNs), a model of which is used in this study, calcium from NMDA receptors (NMDARs) triggers LTP when followed by activation of D₁ dopamine receptors (D₁Rs), while calcium from voltage-gated L-type channels (Ca_v1.3) triggers LTD if paired with inactivation of D₁Rs (Shen et al., 2008 [↗](#); Yagishita et al., 2014 [↗](#); Fisher et al., 2017 [↗](#); Fino et al., 2010 [↗](#); Shindou et al., 2011 [↗](#); Iino et al., 2020 [↗](#)). A threshold for eliciting LTD in SPNs has been experimentally quantified (Shindou et al., 2011 [↗](#)). Although a similar threshold for LTP in SPNs has not been reported yet, presumably such a threshold exists, because weaker stimulation protocols do not induce LTP (Fino et al., 2005 [↗](#)). (However, probably due to the local striatal microcircuitry, a range of different stimulation protocols can induce a range of different plasticity outcomes, as reviewed in Reynolds and Wickens (2002 [↗](#)). An LTP threshold for calcium has been quantified in hippocampal CA1 neurons (Cormier et al., 2001 [↗](#)).

By regulating when plasticity can be induced, metaplasticity has been proposed to prepare neural circuits and networks for learning, stabilize synaptic weights (prevent them from reaching very high or very low values) and to contribute to cognitive effects of disease when it is dysregulated (Hulme et al., 2013 [↗](#)). All metaplasticity models so far have only one modifiable threshold, and typically, synaptic activity above the threshold induces LTP, while activity below the threshold induces LTD (Yger and Gilson, 2015 [↗](#); Bienenstock et al., 1982a [↗](#); Benuskova and Abraham, 2006 [↗](#)). Using a multicompartment model of an SPN, in this article we study what happens when there are two separate calcium thresholds for LTP and LTD, and what their role is in learning. We use a simple learning rule that models plasticity of the dSPN cortico-striatal synapse, to which we add metaplasticity in the two thresholds (Fig. 1A [↗](#)), and apply it to both the linear and nonlinear feature binding problem (FBP and NFBP). The results show that metaplasticity in both thresholds is necessary for successful learning of both tasks, with a specific, opposite and complementary role of each threshold.

Before presenting the results, we introduce the FBP and NFBP. The FBP refers to combining (binding) an object's individual features, such as color, shape or motion, into a unified object (Roskies, 1999 [↗](#); Treisman, 1999 [↗](#); von der Malsburg, 1999 [↗](#); Hardcastle, 2017 [↗](#)). Feature binding is a problem because features such as color, shape and motion, are each processed in different brain regions, and possibly at different times, and need to be combined to represent/construct the perception of the object they correspond to. By applying the simple plasticity and metaplasticity rule to the FBP and NFBP, we in fact study whether a single SPN can perform feature binding, and focus on the role of metaplasticity in learning. In its simplest form, the linear feature binding problem (FBP) consists of discriminating between two patterns, meaning that a neuron should spike when activated by one of them, and remain silent when activated by the other (Fig. 1B [↗](#), top). Each pattern consists of two features, e.g. shape and color, and one of the features can have two values (e.g. 'red' and 'yellow' for the color feature, as opposed to just 'strawberry' for the shape feature in Fig. 1B [↗](#)). Each feature excites the neuron equally, and the term "linear" refers to the computational complexity of the problem, meaning that a neuron able to perform simple linear summation (integration) of incoming stimuli, such as synaptic potentials summated (added) at the soma, can solve the problem. The nonlinear feature binding problem (NFBP) consists of discriminating between four patterns, so that the neuron spikes for two patterns and remains silent for the other two (Fig. 1B [↗](#), bottom). Each pattern also consists of two features, and in the NFBP both features can have two values ('strawberry' or 'banana' for shape, and 'red' or 'yellow' for color, shown in the second table in Fig. 1B [↗](#)). Again, each feature excites the neuron equally, and the additional layer of complexity comes from the requirement to spike for two different patterns whose features do not overlap ('red strawberry' and 'yellow banana'). One of the theoretical solutions to the NFBP requires supralinear integration

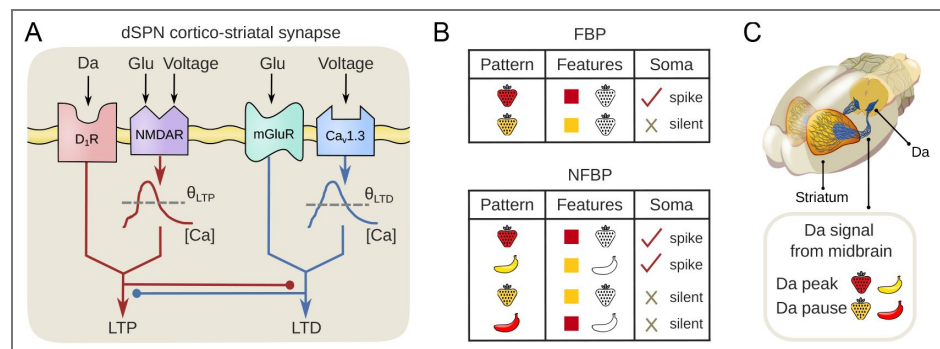


Figure 1. Synaptic plasticity of the cortico-striatal synapse, the tasks for learning, and the dopamine signals for the two types of patterns in the tasks.

(A) Scheme describing the requirements for synaptic plasticity at the cortico-striatal synapse (adapted from Shen et al. (2008) [\[1\]](#) by adding the calcium thresholds, which are the focus of this study). (B) The FBP and the NFBP, illustrated using features common to the visual system. SPNs might instead receive, e.g. sensory- and motor-related features. (C) Dopamine signals from the midbrain are assumed to arrive in the striatum after every pattern. Dopamine peaks are emitted after the relevant patterns, and dopamine pauses after the irrelevant patterns. (The midbrain and its projections are not explicitly modeled in this study, dopamine is programatically provided to the SPN a certain time after pattern arrival.) This figure is adapted from Figure 1 from Khodadadi et al. (2025) [\[2\]](#).

of inputs (Tran-Van-Minh et al., 2015). SPNs exhibit such supralinearities in the form of prolonged dendritic voltage elevations called plateau potentials, demonstrated in the next section. (Supralinear integration of inputs can also solve the simpler FBP.)

As mentioned above, in addition to calcium influx, dopamine signals are also required for plasticity in SPNs. Because of this, we assume that after a pattern is presented, dopamine signals are emitted from the midbrain, and they indicate whether the pattern was “rewarding” or not (Fig. 1C). The patterns for which the SPN should learn to spike are called relevant patterns, and those for which it should be silent are called irrelevant patterns. We assume that dopamine peaks are emitted for relevant patterns, and dopamine pauses for irrelevant patterns (Fig. 1C). After stimulation of the synapse, if the required calcium signal has reached its corresponding threshold, the dopamine signal determines the direction of plasticity. Dopamine peaks are assumed to activate the D_1 Rs, enabling LTP, and dopamine pauses are assumed to deplete dopamine enough so that D_1 Rs are not activated, enabling LTD.

Results

Linear and supralinear integration of inputs by SPNs and corresponding calcium elevations

Figure 2 compares linear integration of synaptic inputs distributed across the dendrites with supra-linear integration of clustered synaptic inputs on one dendrite (Fig. 2A). The number of synaptic inputs is varied from 1 to 30 synapses, and the evoked somatic and spine voltage, as well as the spine $[Ca]_{NMDA}$ and $[Ca]_{L-type}$ are shown in Figs. 2B and 2C. A comparison of the amplitudes of these four quantities between linear and supralinear integration is shown in Fig. 2D. With linear integration, the somatic voltage amplitude increases linearly with the number of synapses (Fig. 2B₁ and blue line in Fig. 2D₁). Conversely, the spine voltage amplitude remains approximately the same, because spines are distributed across the dendrites and are activated in the same way (Fig. 2B₂ and blue line in Fig. 2D₂). The similar amplitudes of the spines’ voltage result in similar $[Ca]_{NMDA}$ and $[Ca]_{L-type}$ elevations (Figs. 2B₃, B₄ and blue lines in Figs. 2D₃, D₄).

On the other hand, when synapses are clustered closely together on a dendrite, SPNs can perform supralinear integration of synaptic inputs through plateau potentials (Du et al., 2017; Plotkin et al., 2011). Plateau potentials are evoked by activating synaptic NMDARs, and, in this study, in addition by glutamate spillover which activates extrasynaptic NMDARs. We included glutamate spillover because it robustly provides an all-or-none jump in the (somatic) voltage amplitude (Trpevski et al., 2023). This all-or-none supralinearity is important for solving the NFBP (Tran-Van-Minh et al., 2015). Fig. 2C₁ shows the somatic voltage elevations when varying the size of the synaptic cluster from 1 to 30 synapses, while Fig. 2C₂ shows the spine voltage in one spine of the synaptic cluster. Similarly, Figs. 2C₃, C₄ show the spine $[Ca]_{NMDA}$ and $[Ca]_{L-type}$ that result from stimulating the synaptic cluster. As explained in the Methods section, calcium from these two different sources is tracked separately in the SPN model. Also, $[Ca]_{NMDA}$ diffuses in the neuron, while $[Ca]_{L-type}$ does not, since it is assumed to be localized in a microdomain. As the cluster size is increased, increasingly larger and longer voltage elevations are produced, as a result of activating more NMDARs (Figs. 2C₁, C₂, traces for cluster sizes below 20 synapses). Once glutamate spillover occurs (here at a cluster size of 20 synapses), an all-or-none plateau potential appears, causing an all-or-none jump in the calcium concentration, as well (Figs. 2C₁-C₄).

This all-or-none behavior is also visible as a jump in the amplitudes of the somatic and spine voltage, and the spine $[Ca]_{NMDA}$ and $[Ca]_{L-type}$ in Figs. 2D₁-D₄. The somatic and spine plateau amplitude shows roughly linear increases before and after glutamate spillover, with the largest non-linearity being caused by the plateau (Figs. 2D₁-D₂, red lines). The plateau also causes a jump in $[Ca]_{NMDA}$ and $[Ca]_{L-type}$, as well (Figs. 2D₃-D₄, red lines). The jump in all four quantities in Fig. 2D is an indication of a “threshold supralinearity”. When a threshold level of stimulation is achieved, the neuron’s somatic voltage amplitude is suddenly much larger, despite it being stimulated with only one additional synapse in the synaptic cluster.

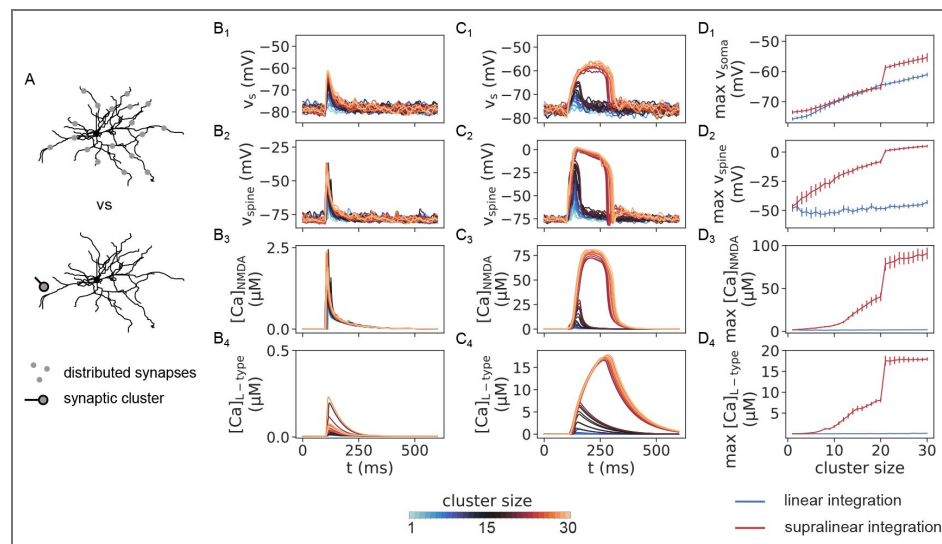


Figure 2. Linear versus supralinear integration of synaptic inputs by SPNs.

(A) Illustrations of the two scenarios of linear integration of distributed synaptic inputs by the soma, and supralinear integration of clustered synaptic inputs in a dendrite. In both scenarios, each synapse is activated with 3 incoming spikes within a window of 35 milliseconds. (B₁-B₄) Somatic voltage, spine voltage, spine [Ca]_{NMDA} and spine [Ca]_{L-type} evoked in the linear integration scenario (randomly distributed across the dendrites). Line color indicates the number of distributed synapses. (C₁-C₄) Somatic voltage, spine voltage, spine [Ca]_{NMDA} and spine [Ca]_{L-type} evoked in the supralinear integration scenario (synaptic cluster placed on one dendrite, in a 20-micrometer region approximately 120 micrometers away from the soma). Similarly, line color represents the size of the synaptic cluster, which is varied from 1 to 30 synapses. (D₁-D₄) The amplitude of the somatic voltage, spine voltage, spine [Ca]_{NMDA} and [Ca]_{L-type} compared between the cases of linear and supralinear integration. Results are averages over 20 trials, and in the case of synaptic clusters, over 8 different dendrites (clusters located in a 20-micrometer in a dendritic region starting at approximately between 120 micrometers from the soma).

Also, the comparison of linear and supralinear integration in Fig. 2D₁ shows that, once a plateau appears, the same number of synapses trigger a much larger somatic voltage elevation when they are clustered compared to when they are randomly distributed across the dendrites. (It also shows a small difference between linear and supralinear integration when the number of stimulated synapses is less than 10. This arises because the synaptic clusters are always placed at approximately the same distance from the soma, while the distributed synapses can be randomly placed further away.)

Synaptic plasticity rule with metaplasticity solves both the FBP and NFBP

The synaptic plasticity rule is simple, and is given with Eq. 1. The synaptic weight w is increased if, during pattern presentation, $[Ca]_{NMDA}$ reaches a level above the LTP threshold, θ_{LTP} , and a dopamine peak arrives afterwards. Similarly, the weight is decreased if $[Ca]_{L-type}$ goes above the LTD threshold, θ_{LTD} , and a dopamine pause arrives afterwards. Synapses can be increased to a maximal level of w_{max} , and decreased to a minimal level of w_{min} , and weight updates are “multiplicative”, i.e. proportional to the current weight value. This formulation is very similar to existing plasticity rules (Rubin et al., 2001; Kistler, 2002; Gütiq et al., 2003). The weight updates are also proportional to a learning rate η , and the time step for advancing the simulation, Δt . This rule is local, meaning that it depends on calcium levels locally at the synapse, and somatic spiking is not necessary for plasticity (similarly to Khodadadi et al. (2025)). This is beneficial for distal synapses, as backpropagating action potentials in general do not affect the local voltage which triggers calcium influx at distant dendritic locations. Also, at least in hippocampal CA3 pyramidal neurons, it has been reported that local dendritic depolarizations are necessary for plasticity, and somatic spiking is not, consistent with the fact that local calcium levels, driven by local voltage, are the signals triggering plasticity (Brandalise et al., 2016).

$$\Delta w = \eta \begin{cases} (w_{max} - w)\Delta t, & [Ca^{2+}]_{NMDA} > \theta_{LTP} \text{ and } Da \uparrow \\ (w - w_{min})\Delta t, & [Ca^{2+}]_{L-type} > \theta_{LTD} \text{ and } Da \downarrow \\ 0, & \text{no Da feedback} \end{cases} \quad (1)$$

Metaplasticity is also implemented in a simple way, given with Eq. 2. It only occurs if synaptic plasticity also occurs. During one update step, each threshold moves closer to the amplitude (maximum) of the calcium level evoked by an arriving pattern, with a rate η_θ (θ_{LTP} follows the amplitude of $[Ca]_{NMDA}$, and θ_{LTD} follows the amplitude of $[Ca]_{L-type}$). This implementation is very similar to the sliding threshold in the well-studied BCM learning rule, where the threshold follows a quantity derived from local synaptic activity (Bienenstock et al., 1982a). When a synapse is modified, both thresholds are updated.

$$\begin{aligned} \Delta \theta_{LTP} &= \eta_\theta ([Ca^{2+}]_{NMDA}^{max} - \theta_{LTP})\Delta t, \\ \Delta \theta_{LTD} &= \eta_\theta ([Ca^{2+}]_{L-type}^{max} - \theta_{LTD})\Delta t. \end{aligned} \quad (2)$$

When using supralinear integration of inputs, we have added an upper threshold for LTP, Θ_{LTP} (Eq. 3). This threshold serves to stop synaptic strengthening once synapses are strong enough to evoke a plateau. With it, synapses are not strengthened to the maximal value, w_{max} , thus avoiding strong supralinear integration for the irrelevant patterns. (This is important for solving the NFBP, as will be seen below.) The upper threshold for LTP only plays a role in synaptic plasticity, and is not taken into account in metaplasticity. This means that metaplasticity is performed in the same way as in Eq. 2, and under the conditions in Eq. 1.

$$\Delta w = \eta \begin{cases} (w_{max} - w)\Delta t, & \theta_{LTP} < [Ca^{2+}]_{NMDA} < \Theta_{LTP} \text{ and } Da \uparrow \\ (w - w_{min})\Delta t, & [Ca^{2+}]_{L-type} > \theta_{LTD} \text{ and } Da \downarrow \\ 0, & \text{no Da feedback} \end{cases} \quad (3)$$

In summary, each synapse is updated based on its local calcium levels, and has its own LTP and LTD thresholds. During learning, patterns in the FBP and NFBP are presented in random order, and after some time, each pattern is followed by a dopamine feedback signal (a dopamine peak or pause), during which plasticity occurs (Fig. 3A). The learning rule is always active (always “on”), avoiding separate training and testing phases sometimes used in other plasticity rules.

Figure 3 shows the outcome of learning on the FBP and NFBP. For the FBP we have tested both linear integration of randomly distributed synaptic inputs (Figs. 3B₁-D₁) and supralinear integration of clustered synapses (Figs. 3B₂-D₂), while for the NFBP we tested only supralinear integration of clustered synapses in two different dendrites (Figs. 3B₃-D₃), since only supralinear integration can solve the NFBP (Tran-Van-Minh et al., 2015). Figs. 3C₁-C₃ show the voltage before and after learning for each pattern in the three scenarios. The SPN is silent before learning, and at the end of learning it has learned to spike to the relevant pattern(s). With supralinear integration, it does this by learning to evoke plateau potentials for the relevant pattern(s). For example, after learning (black traces), ‘red strawberry’ in Figs. 3C₂ and 3C₃ elicits a plateau in dendrite 1, which triggers somatic spiking. For the NFBP, the two relevant patterns evoke plateaus on separate dendrites (‘red strawberry’ on dendrite 1 and ‘yellow banana’ on dendrite 2, Fig. 3C₃). Figs. 3D₁, D₂ show the performance on the FBP with linear and supralinear integration averaged over 50 trials, indicating that both modes of integration can solve the task. For the NFBP, in addition to the input configuration in Fig. 3B₃, with three features on two dendrites, we also tested all possible configurations of four features on two dendrites which allow storage of the relevant patterns on separate dendrites. (These input configurations are listed in Figure 3-figure supplement 1.) Fig. 3D₃ shows that when up to three features are present in one dendrite, the NFBP is successfully solved. When four features are present in one or both dendrites, usually only one relevant pattern is learned. Solving the NFBP in this case requires additional mechanisms such as branch plasticity, as suggested by another study with more abstract neuron models and plasticity rules (Legenstein and Maass, 2011).

Before separately examining the effects of each calcium threshold on learning, it is first necessary to understand how the learning rule successfully solves the tasks by modifying the synaptic weights and thresholds. Figure 4 shows how the synaptic weights and the two calcium thresholds change during learning: they show the same qualitative behavior for both the FBP and NFBP. For the FBP, the synapses carrying the features ‘red’ and ‘strawberry’, which together form the relevant pattern, are strengthened (Figs. 4B₁, B₂). The synapses for ‘yellow’ only weaken, since they are only activated as part of the irrelevant pattern (which is followed by a dopamine pause, a signal for LTD). For the NFBP, where each relevant pattern is stored in a separate dendrite, the outcome is similar: the synapses for ‘red’ and ‘strawberry’ on dendrite 1 and the synapses for ‘yellow’ and ‘banana’ on dendrite 2 are strengthened (Figs. 4B₃, B₄). The remaining synaptic cluster, belonging to one irrelevant pattern in each dendrite, is weakened (the synapses for ‘yellow’ on dendrite 1 and the synapses for ‘strawberry’ on dendrite 2). In all cases, weakening stops when the evoked $[Ca]_{L-type}$ falls below the LTD threshold. For the FBP with linear integration at the soma, many of the randomly distributed synapses for the relevant pattern are strengthened almost to the maximal value w_{max} (Fig. 4B₁). On the other hand, for the FBP with supralinear integration and the NFBP, the clustered synapses for the relevant pattern(s) are not strengthened to their maximal values. Instead, they stabilize once they are strong enough to evoke a plateau (Figs. 4B₂-B₄). This is due to the upper threshold for LTP, Θ_{LTP} , shown with dashed lines in Figs. 4C₂-C₄. (The stabilization takes about twice as long in the NFBP simply because there are twice as many patterns in the task.) When plateaus are evoked by the relevant patterns, they drive somatic spiking (as indicated above in Figs. 3C₂, C₃). An irrelevant pattern stimulates only half of the strengthened synapses on one dendrite. This is not enough for a plateau potential, and even though a supralinear voltage elevation is elicited, no somatic spiking is triggered (Fig. 3C₂, ‘yellow strawberry’ on dendrite 1 for the FBP and Fig. 3C₃, ‘yellow strawberry’ on dendrite 1 and ‘red banana’ on dendrite 2 for the NFBP). Nevertheless, these voltage elevations produce high calcium of both types, but, as will be seen below, the synapses involved are protected from weakening as a result of metaplasticity.

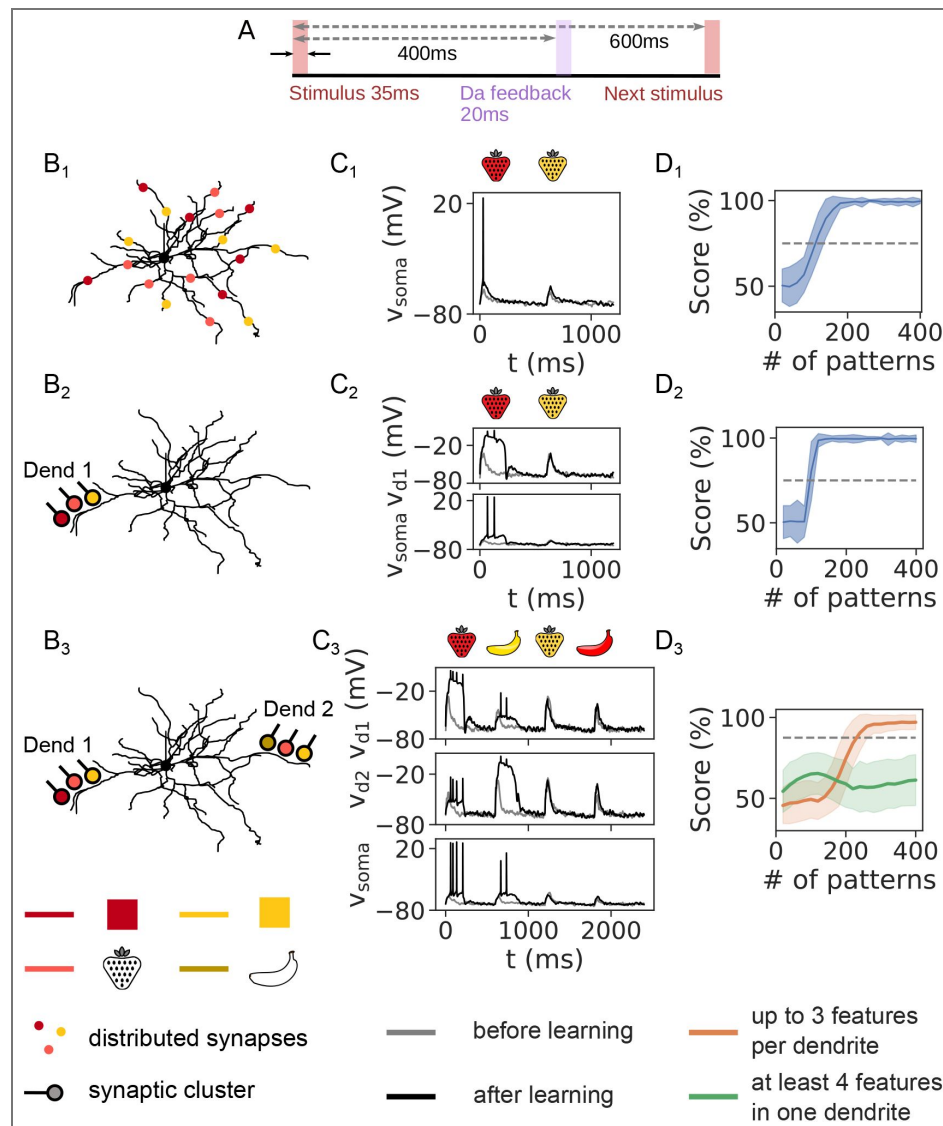


Figure 3. Learning outcome on the FBP and NFBP.

(A) The stimulation protocol used. Patterns are presented in random order. A pattern is represented with synaptic inputs arriving within a window of 35 ms, 3 spikes per synapse. The dopamine signal, during which synapses are updated, is given 400 ms after the start of a pattern, and lasts for 20 ms. The next pattern arrives 600 ms after the start of the current pattern. (B₁) Illustration of the setup for the FBP with linear integration. Each feature is represented by 15 synapses distributed across the dendrites. (B₂) Illustration of the setup for the FBP with supralinear integration. Each feature is represented by 10 synapses clustered in one dendrite. (B₃) Illustration of the setup for the NFBP (only supralinear integration). Each feature is represented by 10 clustered synapses, and clusters are located on two different dendrites. (C₁-C₃) The somatic and dendritic voltages in the FBP with linear (C₁) and supralinear (C₂) integration, and the NFBP (C₃), before and after learning (gray and black traces, respectively), elicited by the relevant and irrelevant pattern(s). (D₁-D₃) The performance on the FBP with linear (D₁) and supralinear (D₂) integration, and on the NFBP (D₃). In (D₁) and (D₂) the performance is averaged over 50 trials, and in (D₂) the cluster was placed at random in one of 8 dendrites in each trial. In (D₃) the performance is averaged over 12 trials per input configuration, which are divided into two groups (with 18 and 13 input configurations, shown in Figure 3—figure supplement 1). (This totals to 216 and 156 trials in the two groups, respectively.) Each point in the plots is the average score over the 20 previous patterns, and the shaded area shows the standard deviation. The dashed lines are the threshold scores for solving the FBP and NFBP, which are 75% and 87.5%, respectively. (See the Methods section for what these threshold scores represent). Figure 3—figure supplement 1. All input configurations used in the NFBP.

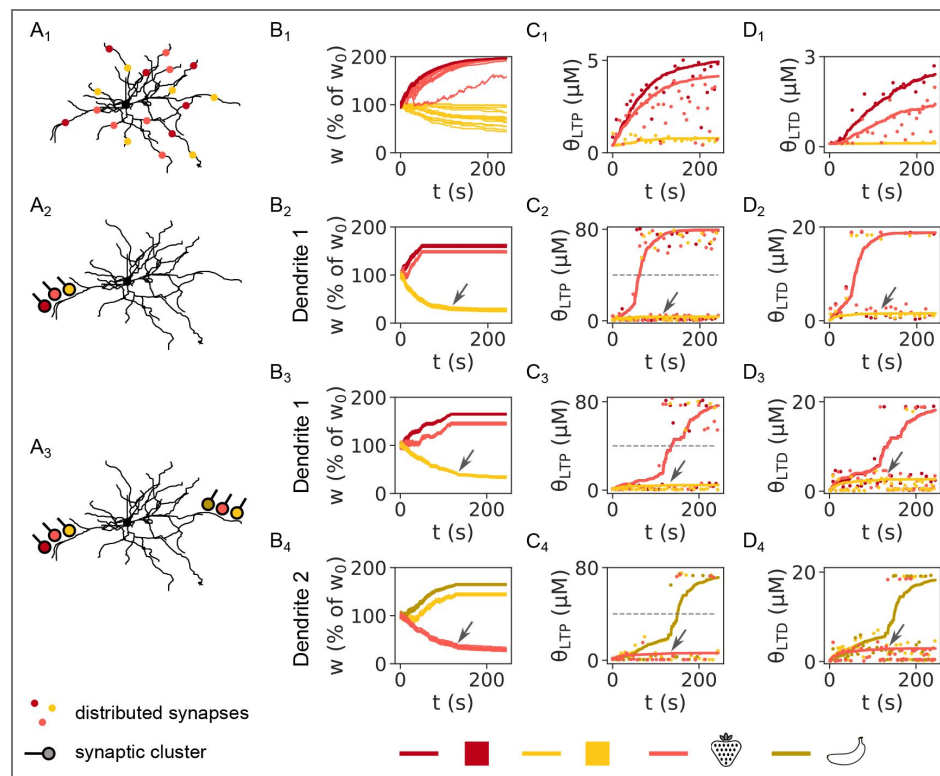


Figure 4. The synaptic weights and calcium thresholds during learning of the FBP and NFBP.

(A) Illustrations of the setups for the FBP with linear (A_1) and supralinear (A_2) integration, and the NFBP (A_3) (same as in Fig. 3). (B-D) The evolution of synaptic weights (B), LTP thresholds (C) and LTD thresholds (D) in the FBP with linear (B_1 - D_1) and supralinear (B_2 - D_2) integration, and the NFBP (B_3 - D_3 and B_4 - D_4). In (B) all synaptic weights for the features are shown, while in (C) and (D) only one synapse per feature is chosen to show its calcium threshold (solid lines). Dots in (C) and (D) represent the amplitudes of $[Ca]_{NMDA}$ and $[Ca]_{L-type}$, respectively, during pattern presentation. For clarity, the calcium amplitudes for some patterns are omitted. (C) and (D) are shown in larger scale in Figure 4-figure supplement 1, with much more detail in the calcium amplitudes. Arrows show the moment in time when the weakened synapses become low enough for their calcium levels to mostly stay below the calcium thresholds. Figure 4-figure supplement 1. Calcium thresholds and calcium amplitudes from Fig. 4 in greater detail. Figure 4-figure supplement 2. The effect of having no upper LTP threshold on supralinear integration. Figure 4-figure supplement 3. The effect of the learning rate on learning. Figure 4-figure supplement 4. The effect of the metaplasticity rate on learning.

The LTP and LTD thresholds for just one synapse from each feature are shown in Figs. 4C [4C](#) and 4D [4D](#), respectively (the same plots are shown with larger panels in Figure 4-figure supplement 1 [1](#) for better clarity). As mentioned above, they show the same behavior for both the FBP and NFBP, the only difference being that they stabilize at different values due to the different calcium levels evoked by distributed and clustered synaptic inputs. Also, the LTP and LTD thresholds show the same behavior when compared to each other. For example, looking at the strengthened synapses first, they increase to follow the calcium amplitudes evoked by each pattern presentation, and stabilize at these values towards the end of the simulation (thresholds for ‘red’ and ‘strawberry’ tend to the dots of the same color above them in Figs. 4C₁-C₃ [4C₁-C₃](#) and 4D₁-D₃ [4D₁-D₃](#), and the thresholds for ‘yellow’ and ‘banana’ tend to the dots of the same color above them in Figs. 4C₄, D₄ [4C₄, D₄](#); this is better visible in Figure 4-figure supplement 1 [1](#) and also highlighted with the light blue regions in Figure 4-figure supplement 1B₃ [1B₃](#)). In the case of linear integration they stabilize at low values, because distributed synapses do not have cooperative voltage effects (due to the large distances between them), and low calcium levels are evoked as a result (Figs. 4C₁, D₁ [4C₁, D₁](#)). Instead, with supralinear integration, the plateaus evoked by the clustered synapses cause much larger voltage and calcium elevations, so the thresholds rise to much higher levels (Figs. 4C₂-C₄ [4C₂-C₄](#) and 4D₂-D₄ [4D₂-D₄](#)).

We next look at the thresholds for the weakened synapses (‘yellow’ synapse in Figs. 4C₁-C₃ [4C₁-C₃](#) and 4D₁-D₃ [4D₁-D₃](#) and ‘strawberry’ synapse in Figs. 4C₄, D₄ [4C₄, D₄](#)). At the beginning, the thresholds increase. Later, once the synapses are weakened enough so that the evoked $[Ca]_{L-type}$ falls below the LTD threshold, they reach a stable level. For example, in Figs. 4C₂, C₃ [4C₂, C₃](#) and 4D₂, D₃ [4D₂, D₃](#), at the beginning of learning, yellow dots can be seen above the ‘yellow’ thresholds, while later they almost always remain below the thresholds. Similarly, in Figs. 4C₄, D₄ [4C₄, D₄](#), light red dots are at first seen above the ‘strawberry’ thresholds, while later they are found below them. This transition of the $[Ca]_{L-type}$ amplitudes from above to below the LTD thresholds is marked with the arrows in Figs. 4B₂-4D₂ [4B₂-4D₂](#), corresponding to the moment in time where synapses are weakened enough. (This is more clearly visible in Figure 4-figure supplement 1 [1](#), and it is also highlighted with the green and brown regions in Figure 4-figure supplement 1B₃ [1B₃](#), and the pink region where yellow dots are absent.) We mention that in the FBP, where the weakened synapses are only activated by the irrelevant pattern, both of their thresholds are only updated during LTD. On the other hand, in the NFBP the weakened synapses and their thresholds can be updated during both LTP and LTD, because these synapses are also activated by a relevant pattern. (However, in Fig. 4 [4](#) this only happens at the very beginning, when the weakened synapses also undergo LTP. As they are being weakened, their LTP threshold is raised, so $[Ca]_{NMDA}$ amplitudes later do not go above the LTP threshold.) In summary, once a calcium amplitude falls below its threshold, one of the necessary conditions to trigger plasticity and metaplasticity is not met, and the thresholds, as well as the synapses, are not updated anymore. This will be much more evident in the next section, where we study the role of metaplasticity.

We stress once again that with supralinear integration, one of the strengthened synaptic clusters in each dendrite is activated by both a relevant and an irrelevant pattern (the ‘strawberry’ cluster in Figs. 4B₂, B₃ [4B₂, B₃](#) and the ‘yellow’ cluster in Fig. 4B₄ [4B₄](#)). Importantly, the LTD threshold for these synapses is raised because of the plateaus (Figs. 4D₂-D₄ [4D₂-D₄](#): ‘strawberry’ synapse in Figs. 4D₂, D₃ [4D₂, D₃](#) and ‘yellow’ synapse in Fig. 4D₄ [4D₄](#)). Because of this, the high $[Ca]_{L-type}$ levels in these synapses evoked from an irrelevant pattern are below the LTD threshold and are not enough to weaken them (e.g. light red dots in Figs. 4D₂, D₃ [4D₂, D₃](#) just above the ‘yellow’ threshold after 100 s are far below their ‘strawberry’ threshold; similarly yellow dots in Fig. 4D₄ [4D₄](#) located just above the ‘strawberry’ threshold after 100s are far below their ‘yellow’ threshold; dots also highlighted in the pink region of Figure 4-figure supplement 1B₃ [1B₃](#)). In this way, these strengthened synapses are protected from weakening, allowing them to store one part of the relevant pattern. This effect will also be much more evident in the next sections, where we look at the effects of metaplasticity on learning.

Briefly, we have also examined the effects of parameters in the learning rule. In supralinear integration, the upper plasticity threshold Θ_{LTP} prevents the strengthened synapses to saturate at their maximal levels. Without it, the FBP is still solved, even though weights attain their maximal values (Figure 4-figure supplements 2B₁ and 2E₁). It is successfully solved because the features innervate only one dendrite, and the saturated weights do not cause somatic spiking when activated by the irrelevant pattern (Figure 4-figure supplement 2A₁). However, the NFBP is not solved without an upper threshold Θ_{LTP} , because the features innervate two dendrites, and the maximally strengthened synapses cause supralinear voltage elevations in the dendrites for the irrelevant patterns, which trigger somatic spiking (Figure 4-figure supplements 2A₂, 2B₂, 2B₃ and 2E₂). The maximally strengthened synapses also drive more weakening in the weakened synapses, compared to when an upper threshold is present (Figure 4-figure supplement 2B). This is due to the elevated $[Ca]_{L-type}$ evoked when an irrelevant pattern activates the maximally strengthened synapses ('strawberry synapses in Figure 4-figure supplements 2B₁, B₂ and 'yellow' synapses in Figure 4-figure supplement 2B₃). The elevated $[Ca]_{L-type}$ for the irrelevant patterns is also visible in the elevated LTD thresholds for the weakened synapses ('yellow' synapse in Figure 4-figure supplement 2D₁, D₂ and 'strawberry' synapse in Figure 4-figure supplement 2D₃, cf. Figs. 4D₂-D₄). (The LTP thresholds for the weakened synapses are also elevated, due to higher $[Ca]_{NMDA}$ produced by the maximally strengthened synapses. This is visible in Figure 4-figure supplement 2C when compared to Figs. 4C₂-C₄)

The learning rate η has an effect on the speed of learning - more pattern presentations are required with lower learning rates, and vice versa (Figure 4-figure supplement 3). The rate of threshold adaptation (metaplasticity rate) η_θ has no significant effect on learning to solve the FBP and NFBP (Figure 4-figure supplement 4).

Metaplasticity in the LTD threshold prevents weakening of strengthened synapses, protecting learned patterns

The role of each threshold is easily examined with this plasticity and metaplasticity learning rule by turning metaplasticity off, i.e. fixing a threshold's value during learning. We focus on the NFBP because it is the harder task, but we also check metaplasticity's effect on the FBP. Fixing the LTD threshold (to the initial low value in the simulation) shows that the synapses encoding the relevant patterns cannot strengthen properly (Figs. 5C₁, C₂). Before we show why this is so, it is important to remember that in the NFBP, a relevant pattern shares one feature with both irrelevant patterns (e.g. 'red strawberry' shares the color feature with 'red banana' and the shape feature with 'yellow strawberry'). Because of this, half of the synapses activated by a relevant pattern are also always activated when an irrelevant pattern arrives.

As learning progresses, the relevant patterns cause strengthening in the synapses that they activate. However, because there is no metaplasticity in the LTD threshold, irrelevant patterns always cause higher $[Ca]_{L-type}$ than the fixed LTD threshold (Figs. 5E₁, E₂ - LTD thresholds stay fixed during learning, and $[Ca]_{L-type}$ is always above θ_{LTD}). This means that when the strengthened synapses are activated by an irrelevant pattern, they will always get weakened. Conversely, when metaplasticity in the LTD threshold is active, it follows the increased levels of $[Ca]_{L-type}$ elicited by a plateau (evoked due to previous synaptic strengthening, as shown previously in Figs. 4D₂-D₄). This elevated LTD threshold protects the strengthened synapses from weakening when they are activated by an irrelevant pattern. As mentioned above, they are protected because an irrelevant pattern, activating only half of the strengthened synapses, does not evoke a plateau, resulting in $[Ca]_{L-type}$ lower than the elevated LTD threshold in Figs. 4D₂ - 4D₄ (e.g. light red dots in pink region in Figure 4-figure supplement 1B₃ are below the elevated 'strawberry' threshold).

Without metaplasticity in the LTD threshold, strengthened synapses are not protected from weakening. As a result, the synaptic weights for the relevant pattern cannot stabilize (Figs. 5C₁, 5C₂), and no plateaus are evoked to trigger somatic spiking (Fig. 5B), so the NFBP cannot be learned (Fig. 5F). The LTP threshold follows the level of $[Ca]_{NMDA}$ and also does not stabilize (Figs. 5D₁, 5D₂). Also, because the LTD threshold is fixed to a low value, the synapses belonging

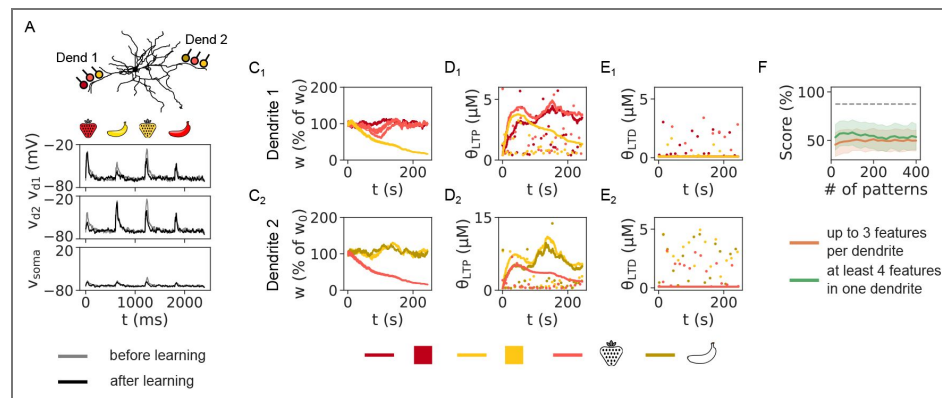


Figure 5. Learning outcome on the NFBP without metaplasticity in the LTD threshold.

(A) An illustration of the setup for the NFBP (same as in Fig. 3B₃). (B) The dendritic and somatic voltage before (gray traces) and after learning (black traces) for all four patterns. (C-E) The evolution of synaptic weights (C), LTP thresholds (D) and LTD thresholds (E) for the synapses in each dendrite. (E) shows the fixed LTD thresholds. In (C) all synapses are shown, and in (D, E) only one synapse per feature is used to show its calcium threshold. Dots represent the amplitudes of $[Ca]_{NMDA}$ (D) and $[Ca]_{L-type}$ (E) during pattern presentation, omitting some patterns for clarity. (F) The performance on the NFBP without metaplasticity in the LTD threshold. Figure 5—figure supplement 1. The effect of a fixed LTD threshold on solving the FBP.

to the weakened feature in each dendrite are almost weakened to the minimal value $w_{\min}$. In summary, by protecting synapses from weakening, the *threshold for weakening* allows proper synaptic *strengthening*.

In the FBP, metaplasticity in the LTD threshold has the same effect. However, due to there being only two patterns in the FBP, the synapses for only one feature in the relevant pattern cannot stabilize (the ‘strawberry’ synapses, Figure 5-figure supplement 1B [↗](#)). This is because the synapses for the feature ‘red’ are only activated by the relevant pattern and, as a result, can only ever be strengthened. Phrased differently, they are never activated by an irrelevant pattern, which would cause them to weaken and be influenced by the fixed LTD threshold. Because the ‘strawberry’ synapses do not stabilize, the clustered synapses in the FBP with supralinear integration are not strong enough to evoke plateaus, and the FBP is not learned (Figure 5-figure supplements 1A₂ [↗](#) and 1E₂ [↗](#)). On the other hand, in the FBP with linear integration, despite these synapses not stabilizing, the strengthening is enough to drive somatic spiking (Figure 5-figure supplement 1A₁ [↗](#)). The FBP with linear integration is solved but with a lower performance (Figure 5-figure supplement 1E₁ [↗](#)).

Metaplasticity in the LTP threshold prevents strengthening of weakened synapses, promoting the weakening of unnecessary synapses

On the other hand, the LTP threshold has an opposite effect on learning. If this threshold is fixed, the feature belonging to the irrelevant pattern in each dendrite cannot be weakened properly (Figs. 6C₁, C₂ [↗](#)). Because of the low value of the LTP threshold, whenever an irrelevant pattern activates these synapses, the evoked $[Ca]_{NMDA}$ is still greater than the threshold (Figs. 6D₁, D₂ [↗](#)), triggering synaptic strengthening even if synapses have been previously weakened. This means that when metaplasticity in the LTP threshold is active, by following the levels of $[Ca]_{NMDA}$ (e.g. LTP threshold for ‘yellow’ in Figs. 4C₁, C₂ [↗](#), and for ‘strawberry’ in Fig. 4C₃ [↗](#)), it prevents these synapses from strengthening when they are activated by the relevant pattern (which can only be stored fully in the other dendrite). As a result, the synapses for ‘yellow’ in Figs. 4B₁, B₂ [↗](#), and for ‘strawberry’ in Fig. 4B₃ [↗](#) are successfully weakened, while in Figs. 6C₁, C₂ [↗](#) they are not. (The LTD threshold behaves as usual for the strengthened synapses, protecting them from weakening, as shown in Figs. 6C₁, C₂ [↗](#) and 6E₁, E₂ [↗](#). However, for the weakened synapses, it does not stabilize.)

Without metaplasticity in the LTP threshold, the NFBP is on average not solved (Fig. 6F [↗](#)). Despite both relevant patterns being stored by each dendrite, the performance is lowered because the SPN sometimes also spikes for the irrelevant patterns (Fig. 6B [↗](#)). Also, in the input configurations containing four features in one dendrite, the average performance even drops below 50 % (green trace in Fig. 6F [↗](#)). In summary, by preventing synapses from strengthening, the *threshold for strengthening* allows proper synaptic *weakening*.

In the FBP, a fixed LTP threshold does not have an effect on learning (Figure 6-figure supplement 1 [↗](#)). The synaptic weights behave the same as with metaplasticity in the LTP threshold being active (Figure 6-figure supplement 1B [↗](#), cf. Figs. 4B₁, B₂ [↗](#)). As a result, the FBP is solved with both linear and supralinear integration (Figure 6-figure supplements 1A [↗](#) and 1E [↗](#)). This is also due to there being only two patterns in the FBP. The weakened ‘yellow’ synapses, whose proper weakening could be affected by the fixed LTP threshold, are only activated during the irrelevant pattern (‘yellow strawberry’). Because they are never activated by the relevant pattern, they can never strengthen, and as a result they remain unaffected by the fixed LTP threshold.

Both thresholds need to be updated during one synaptic modification in order to solve the NFBP

The rule is formulated so that during one synaptic modification (e.g. LTP) the thresholds for both LTP and LTD are updated. Alternatively, it could be formulated so that during LTP, only the threshold for LTP is updated, and during LTD, only that for LTD is updated. We test this variant in

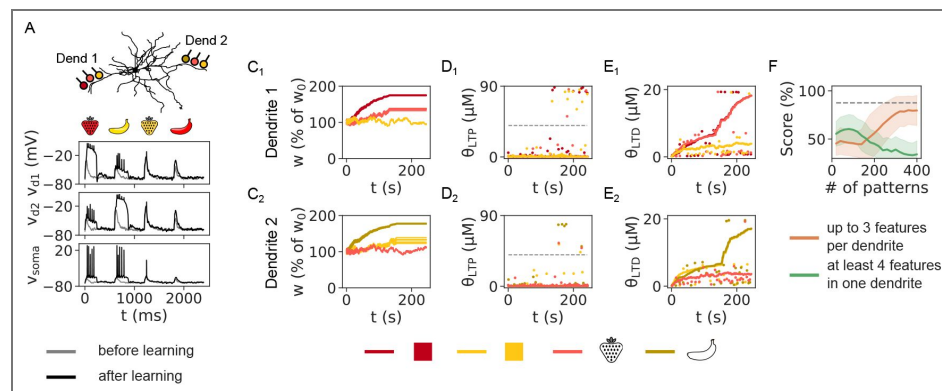


Figure 6. Learning outcome on the NFBP without metaplasticity in the LTP threshold.

(A) An illustration of the setup for the NFBP (same as in Fig. 3B3). (B) The dendritic and somatic voltage before (gray traces) and after learning (black traces) for all four patterns. (C-E) The evolution of synaptic weights (C), LTP thresholds (D) and LTD thresholds (E) for the synapses in each dendrite. (D) shows the fixed LTP thresholds, and the dashed lines show the upper LTP threshold, Θ_{LTP} . In (C) all synapses are shown, and in (D, E) only one synapse per feature is used to show its calcium threshold. Dots represent the amplitudes of $[Ca]_{NMDA}$ (D) and $[Ca]_{L-type}$ (E) during pattern presentation, omitting some patterns for clarity. (F) The performance on the NFBP without metaplasticity in the LTP threshold. Figure 6—figure supplement 1. The effect of a fixed LTP threshold on solving the FBP.

this section, and the results from the previous two sections already give an intuition how this variant would work. Updating only the corresponding threshold during one synaptic modification means that the other threshold is, in effect, kept fixed during that plasticity update, making metaplasticity only partially active. The effects are seen in Figs. 7C₁, C₂. Synapses are neither properly strength-ened, nor weakened, at least within the length of the simulation. For example, one of the relevant patterns ('red strawberry') shows a trend of strengthening, but is also being weakened (Fig. 7C₁). This is because as the LTD thresholds for 'red' and 'strawberry' move higher (Fig. 7E₁), LTD in these synapses is (sometimes) prevented, and a trend of synaptic strengthening appears. Also, as the LTP threshold for 'yellow' increases (Fig. 7D₁), it prevents these synapses from strengthening, and as a result, they weaken to a degree.

However, since the LTD threshold is not updated during LTP, it does not follow the $[Ca]_{L-type}$ levels generated by the relevant patterns. It is updated only during LTD, following the $[Ca]_{L-type}$ levels from irrelevant patterns. For example, in dendrite 1, the 'red banana' activates only the cluster representing 'red', evoking little $[Ca]_{L-type}$ and, as a result, the LTD threshold for the 'red' synapse is lower (Fig. 7E₁). On the other hand, the 'yellow strawberry' activates two synaptic clusters (Fig. 7A₁), causing a larger voltage elevation and consequently larger $[Ca]_{L-type}$, so the LTD thresholds for these synapses move higher (Fig. 7E₁). However, because of the trend of synaptic strengthening in the 'red' and 'strawberry' synapses, the irrelevant patterns sometimes still evoke $[Ca]_{L-type}$ higher than the LTD thresholds. As a result, the synapses for 'red' and 'strawberry' in dendrite 1 are not protected from weakening (when activated as part of the irrelevant patterns). Also, as the LTD threshold for 'yellow' increases in the first 70 s of the simulation (Fig. 7E₁), the synapses for 'yellow' show a trend of increasing in this period, as well, before being decreased afterwards (when the 'yellow' LTP threshold is increased).

On the other hand, in dendrite 2 the relevant pattern is not strengthened, and it is not clear whether more learning trials will cause both relevant patterns to be stored in the two dendrites. On average, the NFBP is not solved in this case (Fig. 7F).

Finally, this variant is similar to not having metaplasticity at all (Fig. 8). Without metaplasticity, no synapses stabilize, i.e. they are neither strengthened nor weakened (Fig. 8C₁, C₂).

For the FBP, the results in the previous two sections again give an intuition for how partial metaplasticity would work. As shown in Figure 6-figure supplement 1, even when the LTP threshold is fixed, it does not have an effect on solving the FBP. Similarly, neither does it have an effect when it is partially active: the LTP threshold is merely less elevated than compared to when metaplasticity is always active (Figure 7-figure supplement 1C, cf. Figs. 4C₁, C₂). On the other hand, partial meta-plasticity in the LTD threshold is not enough to protect strengthened synapses from weakening. The LTD threshold does not follow $[Ca]_{L-type}$ fast enough (Figure 7-figure supplement 1D), and as a result the 'strawberry' synapses are not strengthened properly (even though, for example, their weights stabilize when using supralinear integration, Figure 7-figure supplement 1B). Similarly to Figure 5-figure supplement 1, this does not affect learning to solve the FBP with linear integration (Figure 7-figure supplement 1A₁ and 1E₁), but it does affect learning of the FBP with supralinear integration (Figure 7-figure supplement 1A₂ and 1E₂).

The effects of having no metaplasticity on the FBP are similar to those of having partial metaplasticity (Figure 8-figure supplement 1). Again, the synapses encoding one of the features in the relevant pattern cannot stabilize (those for 'strawberry', Figure 8-figure supplement 1B). This only moderately affects the performance on the FBP with linear integration (Figure 8-figure supplement 1A₁ and 1E₁), but is necessary for solving the FBP with supralinear integration (Figure 8-figure supplement 1A₂ and 1E₂). In summary, this shows that in the FBP, metaplasticity is important when a feature is represented with clustered synapses. It allows their synaptic weights to stabilize, and to evoke plateau potentials that can drive somatic spiking. Conversely, when a feature activates distributed synapses across the dendrites, the FBP can still be solved without metaplasticity, even if some synapses do not stabilize.

Figure 7. Learning outcome on the NFBP with partial metaplasticity, where the LTP threshold is updated only during LTP, and the LTD threshold is updated only during LTD.

(A) An illustration of the setup for the NFBP (same as in Fig. 3B₃). (B) The dendritic and somatic voltage before (gray traces) and after learning (black traces) for all four patterns. (C-E) The evolution of synaptic weights (C), LTP thresholds (D) and LTD thresholds (E) for the synapses in each dendrite. In (C) all synapses are shown, and in (D, E) only one synapse per feature is used to show its calcium threshold. Dots represent the amplitudes of $[Ca]_{NMDA}$ (D) and $[Ca]_{L-type}$ (E) during pattern presentation, omitting some patterns for clarity. (F) The performance on the NFBP with partial metaplasticity. Figure 7—figure supplement 1. The effect of partial metaplasticity on solving the FBP.

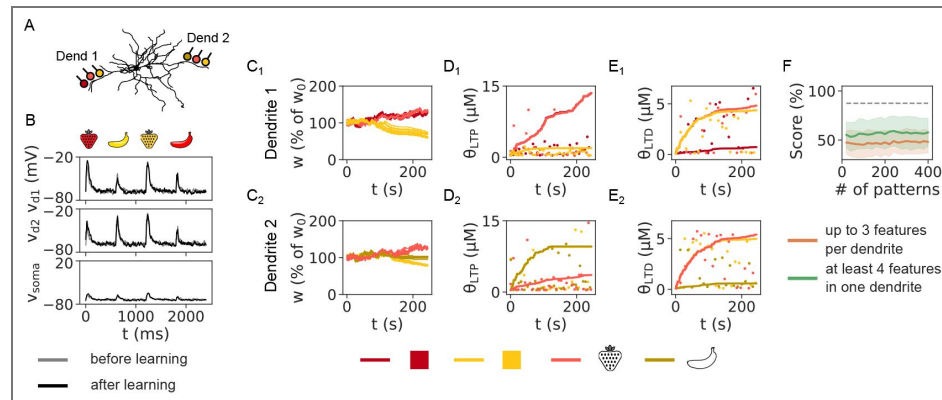
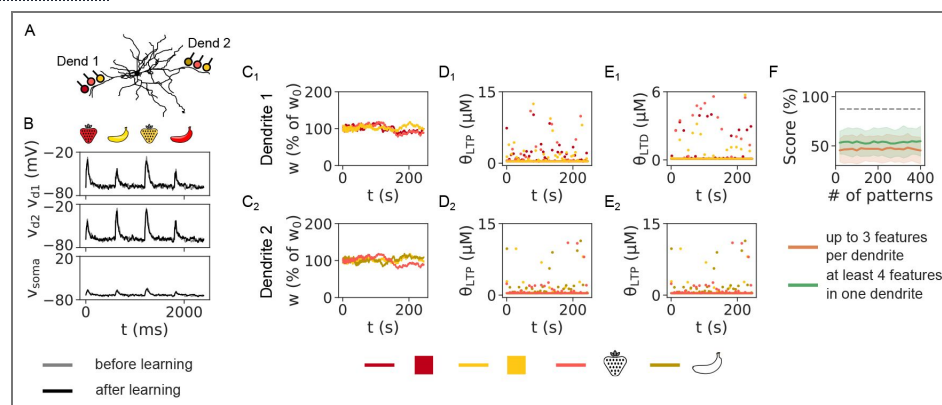


Figure 8. Learning outcome on the NFBP without metaplasticity (the LTP and the LTD thresholds are fixed).

(A) An illustration of the setup for the NFBP (same as in Fig. 3B₃). (B) The dendritic and somatic voltage before (gray traces) and after learning (black traces) for all four patterns. (C) The evolution of synaptic weights in both dendrites. (D, E) The fixed LTP thresholds (D) and LTD thresholds (E) in each dendrite, with calcium amplitudes during pattern presentation shown with dots (omitting some patterns for clarity). (F) The performance on the NFBP without metaplasticity. Figure 8—figure supplement 1. The effect of having no metaplasticity on solving the FBP.



Discussion

In this article we studied the effects of metaplasticity in two separate calcium thresholds for inducing synaptic plasticity, one of which is a threshold for LTP, and the other for LTD. Plasticity occurs when calcium levels from different sources, implicated in synaptic plasticity in SPNs, reach levels above the thresholds. We studied the effects of metaplasticity on how well the SPN learns the FBP and NFBP, tasks which require linear and supralinear summation of synaptic inputs, respectively. The results showed that:

1. Without metaplasticity, the NFBP cannot be solved. It is also necessary for solving the FBP with supralinear integration, but it is not necessary for solving the FBP with linear integration.
2. Metaplasticity in the LTD threshold protects learned patterns from being weakened, allowing the LTP process to properly strengthen (encode) the patterns. In effect, *the threshold for weakening allows proper synaptic strengthening*.
3. Metaplasticity in the LTP threshold prevents unnecessary features from being strengthened, allowing the LTD process to persistently weaken them. In effect, *the threshold for strengthening allows proper synaptic weakening*.
4. For successful learning of the NFBP, metaplasticity in both thresholds needs to occur during one synaptic modification, irrespective of whether the modification is LTP or LTD.

In summary, the two calcium thresholds have opposite and complementary roles in regulating the strengthening and weakening of synapses. The ultimate effects of metaplasticity are unexpected, at least at first glance, since the threshold for one plasticity type (e.g. LTD) allows proper expression of the other plasticity type (e.g. LTP). Metaplasticity is particularly important for clustered synapses, where the cooperative, supralinear voltage elevations can cause large calcium influx - it allows synaptic plasticity to adapt to this large calcium influx. By regulating plasticity in clustered synapses, metaplasticity allows neurons to perform computationally difficult tasks (such as the NFBP) that rely on supralinear integration by the clustered synapses. Simpler tasks that can be solved with linear integration of synaptic inputs (such as the FBP) do not require metaplasticity.

This study also provides a basis for exploring other metaplasticity rules. Here we only considered metaplasticity that is activated as a consequence of synaptic plasticity. However, stimulation that does not cause synaptic plasticity can also activate metaplasticity, affecting the induction of future plasticity (Abraham, 2008 [\[link\]](#)). Also, we only considered metaplasticity evoked in the same synapses activated by presynaptic spikes (homosynaptic metaplasticity). However, heterosynaptic metaplasticity mechanisms also exist, where the evoking stimuli originate from different synapses or different neurons (Abraham, 2008 [\[link\]](#); Abraham et al., 2019 [\[link\]](#); Hulme et al., 2014 [\[link\]](#)). Furthermore, as detailed calcium models for other neurons are developed, the contributions of different calcium sources in triggering synaptic plasticity can be studied using this simple learning rule as a basis. Regarding SPNs specifically, more realistic metaplasticity rules can also be developed once future experimental findings appear.

We have made two assumptions when formulating the learning rule. The first is that metaplasticity operates in cortico-striatal synapses, which is not a strong assumption, since metaplasticity has been reported for many synapses. The second – a major assumption – is the upper threshold for synaptic strengthening, which means that high $[Ca]_{NMDA}$ during LTP conditions (a dopamine peak) prevents synaptic plasticity (blocks LTP). It is an open question whether this assumption is true. Nevertheless, there are some molecular mechanisms which could form the basis for such an upper threshold. In the neuromodulatory branch of the signaling network in Fig. 1 [\[link\]](#) (the branch activated by dopamine), the enzyme adenylyl cyclase 5, which transduces the dopamine signal from the D_1R , is inhibited by calcium. This indicates that high $[Ca]_{NMDA}$ might prevent synaptic strengthening already very early in the signaling network. Further downstream in the signaling network, calcium activates PP2B through calmodulin, and PP2B inactivates PKA, an enzyme necessary for synaptic strengthening (Church et al., 2021 [\[link\]](#); Yagishita et al., 2014 [\[link\]](#)). This is another route through which high $[Ca]_{NMDA}$ could prevent LTP. Due to the complexity of the signaling network, the influence of high $[Ca]_{NMDA}$ on synaptic

plasticity requires further study, and kinetic models of the cortico-striatal synaptic signaling network which incorporate the many molecular interactions might provide some answers (Lindskog et al., 2006 [↗](#); Nakano et al., 2010 [↗](#); Nair et al., 2015 [↗](#)).

In summary, at least when it comes to solving the NFBP, the proposed learning rule highlights the importance of deciding when to stop updating the synaptic weights. The two processes (mechanisms) that regulate when synaptic plasticity occurs are the upper plasticity threshold for LTP and metaplasticity. Without the upper plasticity threshold, all strengthened synapses would reach their maximally attainable synaptic weight. Without metaplasticity, the synaptic weights would not stabilize. Important future questions relate to the various such possible mechanisms and their effects on learning, as it is likely that different types of synapses possess different metaplasticity mechanisms, which support specific functions in various brain regions. Some examples of different metaplasticity mechanisms are known in the hippocampus. In the Schaffer collateral synapses to the CA1 area, prior activation of NMDARs (which does not induce long-term plasticity) temporarily increases the LTP threshold and prevents LTP, while prior activation of mGluRs facilitates LTP (Huang et al., 1992 [↗](#); Cohen and Abraham, 1996 [↗](#)). In the synapses from the perforant path to the dentate gyrus, prior stimulation at theta frequency typical for the hippocampus, lowers the threshold for both LTP and LTD (Christie et al., 1995 [↗](#); Christie and Abraham, 1992 [↗](#)). Additionally, prior stimulation at a high frequency also lowers the LTD threshold in both unchanged and in previously strengthened synapses (Holland and Wagner, 1998 [↗](#)).

Finally, we have recently participated in another study asking whether SPNs could solve the NFBP, using a different learning rule that incorporates more biological complexity/detail (Khodadadi et al., 2025 [↗](#)). That study focused on dendritic integration which was much less supralinear, i.e. the dendrites were in a regime where they produced so-called “boosting” nonlinearities, that are likely common physiologically. Completely solving the NFBP required both excitatory and inhibitory synaptic plasticity, regardless of whether the synaptic clusters were closer or further away from the soma. On the other hand, the all-or-none plateau potentials in this study achieve a very high score on the NFBP with excitatory plasticity alone. This indicates that with a threshold non-linearity, the SPNs might completely solve the task without requiring inhibitory synapses and even for very distally located synaptic clusters. Whether this is true will be explored in another study, as theoretical considerations suggest that the NFBP can be solved with excitatory synapses only (Tran-Van-Minh et al., 2015 [↗](#)).

Methods

Plasticity rule

We highlight again that, similar to Khodadadi et al. (2025 [↗](#)), the learning rule is:

- local: only signals at the local synaptic site are used to trigger plasticity, calcium- and reward-based:
- local calcium levels trigger plasticity, the type of which is decided by the dopamine reward signal, and
- always “on”: no separate training and testing phases are needed because learning is regulated by metaplasticity.

Differently from Khodadadi et al. (2025 [↗](#)), the learning rule has metaplasticity in both calcium thresholds, and is also (much) simpler, making it suitable for studying the thresholds’ roles in synaptic plasticity.

The parameters used in the plasticity rule are given in Table 1 [↗](#). The values of $w_{\max}$ and $w_{\min}$ are not varied throughout the simulations. They describe the maximal and minimal values that the synaptic weight can take on in the model. Whether these values are reached depends on the metaplasticity mechanisms (the values of the calcium thresholds), and, in the case of supralinear integration, the upper plasticity threshold Θ_{LTP} , which is a fixed parameter that can be set at any value in the region of the supralinearity (the jump) in $[\text{Ca}]_{\text{NMDA}}$ in Fig. 2D₃ [↗](#) (we used 40 μM). The

learning rate η and the metaplasticity rate η_θ are also kept fixed throughout all simulations, except in Figure 4–figure supplements 3 and 4, where they are varied to study their effect on learning. Finally, the LTP and LTD thresholds, θ_{LTP} and θ_{LTD} , are initialized to low values, $\theta_{\text{LTP},0}$ and $\theta_{\text{LTD},0}$, which are also the minimal values that these thresholds can take on. They are initialized to low values in order to showcase learning in the synapses. Otherwise, if they are set to high values, they may act to protect synapses from weakening (high $\theta_{\text{LTD},0}$) or prevent them from strengthening (high $\theta_{\text{LTP},0}$), as is shown with Figs. 5 and 6. The synaptic weights are initialized uniformly at random from the interval [0.3, 0.35].

Neuron model

We use a morphologically realistic, multicompartmental model of a direct-pathway striatal projection neuron from the collection of 71 such models in Lindroos and Hellgren Kotaleski (2020) (index 34 in the collection). As detailed in Lindroos and Hellgren Kotaleski (2020), the ionic conductances in all models in the collection have been selected with a search process to produce frequency-current (F-I) responses and dendritic calcium elevations arising from backpropagating action potentials that match experimental data, and in addition produced a suitable number of spikes to injected current, no voltage oscillations leading up to the spikes, proper afterhyperpolarization behavior, and an F-I slope within three standard deviations of the experimentally determined mean.

The main contributor to supralinear integration in the SPN model is the activation of NMDARs. This has been shown, for example, in Fig. S7 in Du et al. (2017), where supralinear integration with clustered synapses is similar in an active and a passive dendrite in the SPN. This means that the NMDA synapses, and not the SPN's ion channels, are causing it. SPN-specific ion channels do have a large effect on the SPN's electrophysiological behavior – the lowered resting potential of around -85 mV is a result of the strongly inwardly rectifying K^+ channels (Nisenbaum and Wilson, 1995; Wilson, 2005). Because of this, to raise their somatic voltage, the SPNs require strong synaptic drive from distributed synaptic inputs across the dendrites, or a localized activation of clustered synapses which generates a plateau potential. With this behavior, SPNs are said to have two states: a “down state”, which requires strong synaptic drive to reach the firing threshold, and an “up state”, where they are more sensitive to incoming synaptic inputs (Plotkin et al., 2011). It has also been suggested that plateau potentials themselves cause the transition from a “down” state to an “up” state (Oikonomou et al., 2014). Plateau potentials in SPNs normally do not cause somatic spiking without additional synaptic inputs, which is why in the simulations we have used elevated background noise.

In this study, synapses are placed on spines, which are not present in the original model in Lindroos and Hellgren Kotaleski (2020). The spines consist of a neck and head with neck and head with lengths and diameters $l_{\text{neck}} = 0.5\mu\text{m}$, $l_{\text{head}} = 0.5\mu\text{m}$, and $d_{\text{neck}} = 0.125\mu\text{m}$, $d_{\text{head}} = 0.5\mu\text{m}$, respectively. Axial resistance in all compartments of the model is $150\ \Omega \cdot \text{cm}$, except for the spine neck, where it is $1130\ \Omega \cdot \text{cm}$ (Dorman et al., 2018). The spines contain the inwardly rectifying K^+ channel, and the same voltage-gated calcium channels as in the dendrites. The spine calcium channel conductances have been manually tuned to match the relative proportions determined in Carter and Sabatini (2004) and Higley and Sabatini (2010), as well as the Ca^{2+} concentration amplitudes arising from backpropagating action potential (bAP) stimulation as in Fig. 2 of Shindou et al. (2011).

Calcium dynamics and diffusion model

Because calcium from two different sources triggers LTP and LTD in SPNs, we tracked calcium from the two sources separately. Calcium from NMDA channels, when followed by a peak in dopamine, triggers LTP (Shen et al., 2008; Yagishita et al., 2014; Fino et al., 2010; Fisher et al., 2017). Calcium from $\text{Ca}_v1.3$ channels, when paired with a pause in dopamine, has been shown to be necessary for LTD (Wang et al., 2006; Fino et al., 2010; Shindou et al., 2011; Iino et al., 2020).

Parameter	Value
$w_{\max}$	2
$w_{\min}$	0.1
$\Theta_{\text{LTP}} \text{ (}\mu\text{M)}$	40
η	0.85
η_{θ}	2
$\theta_{\text{LTP},0} \text{ (}\mu\text{M)}$	0.4
$\theta_{\text{LTP},0} \text{ (}\mu\text{M)}$	0.1

Table 1. The parameters in the plasticity rule.

Without implementing calcium diffusion explicitly, the accumulation of $[Ca]_{NMDA}$ in a spine shows a strange effect when the synaptic cluster size is gradually increased, which is that the largest plateaus do not correspond to the largest calcium amplitude (Figs. 9A, B). This is simply a result of less NMDAR membrane current flowing in as the plateau's voltage approaches the NMDAR reversal potential in Eq. 6. Larger plateaus giving rise to lower $[Ca]_{NMDA}$ amplitudes could pose a problem to the learning rule, since the decision when to trigger plasticity and metaplasticity is made by comparing $[Ca]_{NMDA}$ amplitude against the LTP threshold. In a synaptic cluster that is being strengthened, but starts to evoke a smaller $[Ca]_{NMDA}$ amplitude after some strengthening due to the effect in Figs. 9A, B, learning might be prematurely stopped as the $[Ca]_{NMDA}$ amplitude falls below the increased LTP threshold.

Because of this, we sought to obtain a monotonic increase of calcium amplitude with increasing plateau voltage (as in Fig. 2C), for which it was necessary to implement axial diffusion and buffering for $[Ca]_{NMDA}$. Because tuning a detailed calcium diffusion model to data is a project in its own right, we used the most detailed calcium diffusion model for SPNs available to date (by Dorman et al. (2018)) as a reference. We reuse the same buffer molecules, their amounts and their calcium binding and unbinding rates from Dorman et al. (2018) with small changes, given in Table 2. The rationale for reusing them is that the idealized morphology used in Dorman et al. (2018) should on average represent any other SPN, so the parameters obtained from the optimization procedure in that study should on average also apply to other multicompartment SPN models. The model in Dorman et al. (2018) also includes a plasma-membrane Ca pump (PMCA) in the soma, dendrites and spines, which we added in this model, and a sodium/calcium exchanger in the spines, which we omitted, because with the parameters from Dorman et al. (2018), it did not noticeably alter the spine $[Ca]_{NMDA}$ concentration. In the calcium diffusion model in Dorman et al. (2018), calcium originating from all sources, which in that study are NMDARs and voltage-gated calcium channels (VGCCs), and a minor, 0.1% contribution of the AMPAR current, diffuses throughout the SPN model. Because the calcium buffer parameters are optimized to such a scenario, we also included calcium from the $Ca_v2.1$ (Q-type), $Ca_v2.2$ (N-type), $Ca_v2.3$ (R-type), $Ca_v3.2$ and $Ca_v3.3$ (T-type) channels in the SPN together with $[Ca]_{NMDA}$ in the diffusion model. Indeed, non-selective blockade of voltage-gated calcium channels with mibefradil has shown that they are also necessary for inducing LTP in SPNs (Fino et al., 2010). Mibefradil has been shown to block T-type, L-type, R-type, N-type and P/Q-type calcium channels in different experimental preparations, so, apart from the L-type channels, we included all of them in the diffusion model (Randall and Tsien, 1997; Viana et al., 1997; Aczél et al., 1998; Eller et al., 2000; Leuranguer et al., 2001). What the role of each of these channels is in LTP in SPNs requires further study with selective blockers.

We also show how diffusion of only $[Ca]_{NMDA}$ looks like in the SPN in Figure 9-figure supplement 1 (and Fig. 9C). The difference with the diffusion model used throughout the study, which includes VGCCs, is minor (Figure 9-figure supplement 1, cf. Figs. 2C, D). The qualitative and quantitative shapes of the voltage and calcium elevations are very similar (Figure 9-figure supplement 1A, cf. Fig. 2C). Regarding the amplitudes of these quantities, there is a 1 mV increase in the evoked somatic amplitude after glutamate spillover when only $[Ca]_{NMDA}$ diffuses (Figure 9-figure supplement 1B), and an obvious difference in the amplitude of $[Ca]_{NMDA}$ (Figure 9-figure supplement 1B). Nevertheless, the threshold nonlinearity in the $[Ca]_{NMDA}$ amplitude is still present, and the learning rule would work the same provided the upper plasticity threshold Θ_{LTP} is placed in the region of the nonlinearity.

Even though mibefradil also affects L-type channels, we did not include them in the axial diffusion model together with $[Ca]_{NMDA}$ because L-type calcium channels frequently form calcium microdomains, where calcium entry and diffusion is restricted within a very small, micrometer range (Berridge, 2006; Chen and Sabatini, 2012; Parekh, 2008). In the microdomain, the L-type channels can be physically coupled in supramolecular protein complexes, further contributing to the local effect of the calcium influx in activating the relevant signaling networks, and this is also the case for $Ca_v1.3$ channels in SPNs (Olson et al., 2005; Stanika et al., 2015). Because of this, we assume that such highly localized $[Ca]_{L-type}$ signals trigger LTD in the synapse,

Figure 9. Alternative models of calcium accumulation.

(A) Spine $[Ca]_{NMDA}$ without axial diffusion. $[Ca]_{NMDA}$ is described with a pool model similar to that in Eq. 4 (except that instead of $[Ca]_{L-type}$, this pool accumulates $[Ca]_{NMDA}$ arising from the NMDA current in Eq. 6). (B) Spine $[Ca]_{NMDA} + [Ca]_{VGCC}$ without axial diffusion. Calcium from the NMDA current and all voltage-gated calcium channels (except L-type) accumulates in a pool model. (C) Spine $[Ca]_{NMDA}$ with axial diffusion. Compared to Fig. 2C, in this plot only calcium from NMDARs diffuses in the neuron, without calcium from other voltage-gated calcium channels. Axial diffusion of calcium from NMDARs only.

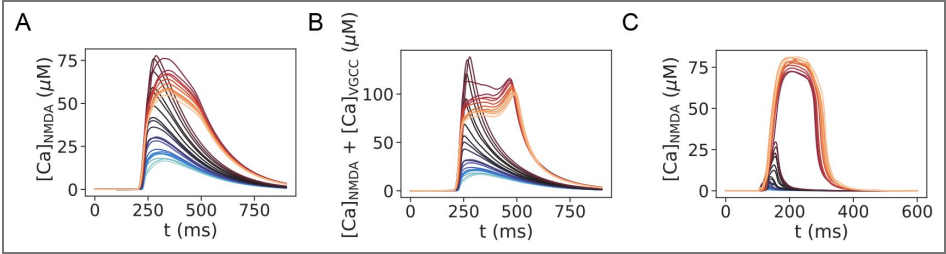


Table 2. The parameters for the axial calcium diffusion model.

Compared to Dorman et al. (2018), we have reduced the concentration of the immobile calcium buffer from 2.5 mM to 0.15 mM and changed the catalytic rate of the PMCA. The diffusion coefficient of Ca^{2+} is $200 \frac{\mu m^2}{s}$ (same as in Dorman et al. (2018)).

Buffers	k_f (mM/ms)	k_r (/ms)	Concentration (mM)	D ($\mu m^2/s$)
Calbindin	28	$19.6 \cdot 10^{-6}$	0.08	66
CaMN	100	10^{-3}	0.02	66
CaMC	6	$9 \cdot 10^{-6}$	0.02	66
Immobile	400	40	2.5	0
Pump	k_f [mM/($\mu m^2/ms$)]	K_d (mM)		
PMCA	5000	0.3		

and calculated $[Ca]_{L-type}$ with a separate, phenomenological pool model, which describes $[Ca]_{L-type}$ in a thin cylindrical shell under the membrane. The implementation follows that of Wolf et al. (2005), according to Eq. 4. In this model $[Ca]_{L-type}$ decays with a time constant τ_{Ca} , modeling Ca^{2+} diffusion only phenomenologically, i.e. no Ca^{2+} ions move between dendritic compartments. Compared to the very simplest pool models, this pool model also includes a Ca^{2+} pump that extrudes intracellular Ca^{2+} to the extracellular space. F is the Faraday constant, k_t is the catalytic activity of the pump, K_d is the pump dissociation constant for Ca^{2+} , d is the depth of the submembrane shell where Ca^{2+} accumulates, and k and p are phenomenological parameters used in Wolf et al. (2005) used to balance Ca^{2+} influx and efflux. The values for the parameters in Eq. 4 are given in Table 3.

$$\frac{d[Ca]_{L-type}}{dt} = \frac{[Ca]_{\infty} - [Ca]_{L-type}}{\tau_{Ca}} + \frac{k}{2Fd} \sum_{x \in \{1.2, 1.3\}} I_{Ca_{v,x}} - \frac{pk_t}{[Ca]_{L-type} + K_d} [Ca]_{L-type} \quad (4)$$

In this pool model we included calcium from both $Ca_v1.2$ and $Ca_v1.3$ channels, since there are no pharmacological agents that selectively block only one of the channels, i.e. the compounds used in the experiments which have determined $Ca_v1.3$ as necessary for LTD in cortico-striatal synapses would also block $Ca_v1.2$ channels in SPNs (Berger and Bartsch (2014); nimodipine in Wang et al. (2006) and Shindou et al. (2011); also, other non-selective L-type channel blockers are used to block LTD in the following studies: nifedipine in Calabresi et al. (1994); nitrendipine in Kreitzer and Malenka (2005); mibefradil in Fino et al. (2010); finally, experiments in Plotkin et al. (2013) have suggested that other VGCCs than $Ca_v1.3$ may be involved in activating calcium-induced calcium release, which is necessary for LTD).

Synaptic inputs

There are two different models of synapses used in this study. One model – the dual exponential model – is used in describing the background noise, which consists of glutamatergic and GABAergic synapses, distributed across the dendrites according to the distribution measured in Cheng et al. (1997). These synapses are fixed (they have no plasticity) because they are activated at a low rate of 1 Hz. Compared to the 35 ms activation window in Fig. 3, this rate is low enough so that we decided to neglect any background noise synapses that might be co-active in the same time window and thus become eligible for plasticity together with the feature-carrying synapses. Because there are many synapses representing the background noise, this assumption also lowers the computational cost – if they were plastic, the learning rule would need to be computed for each of them.

The other model, used to describe the synapses carrying the features, is a saturating synapse model. We chose a saturating synapse model because we activate these synapses with 3 spikes each within a 35 ms window, and wanted to avoid any unrealistically high conductance values that could otherwise arise in non-saturating models.

Dual exponential model

The dual exponential synaptic model (a difference of two exponential functions) for a spike arriving at time 0 is represented by:

$$g_{syn}(t) = g_{max} \frac{1}{C} \frac{\tau_A \tau_B}{\tau_B - \tau_A} \left(\exp\left(-\frac{t-t_0}{\tau_B}\right) - \exp\left(-\frac{t-t_0}{\tau_A}\right) \right) \quad (5)$$

where g_{max} is the maximal synaptic conductance, and τ_A and τ_B are the rise and decay time constants. C is a normalization factor given with:

$$C = \frac{\tau_A \tau_B}{\tau_B - \tau_A} \left(\exp\left(-\frac{t^*}{\tau_B}\right) - \exp\left(-\frac{t^*}{\tau_A}\right) \right)$$

$$t^* = \frac{\tau_A \tau_B}{\tau_B - \tau_A} \ln \frac{\tau_B}{\tau_A}$$

Parameter	Value
$[Ca^{2+}]_{[L-type]}$ (nM)	50
τ_{Ca} (ms)	100
k_i (mM/ms)	10^{-4}
K_d (mM)	10^{-4}
d (μ m)	0.1
k	10^4
p	0.02

Table 3. The parameters for the pool model for $[Ca]_{L-type}$.

where t^* is the position of the maximum of the dual exponential function. This normalization ensures that the time-varying part of the dual exponential reaches a peak value of 1, which is then scaled by the maximal conductance $g_{\max}$. (The implementation in NEURON contains a synaptic weight parameter, as well. Since in the dual exponential model we have fixed the weight to 1, it is omitted from Eq. 5.)

The synaptic current arising from the conductance g_{syn} is:

$$I_{\text{syn}} = g_{\text{syn}} (V - E_{\text{rev}}).$$

Excitatory synapses have both AMPA and NMDA components, and NMDA synapses in addition have a Mg^{2+} block described by a sigmoidal function:

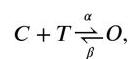
$$I_{\text{NMDA}} = g_{\text{NMDA}} (V - E_{\text{rev}}) \sigma; \sigma = \frac{1}{1 + \eta \exp(-\alpha V)}, \quad (6)$$

with parameters $\alpha = 0.062$ and $\eta = 0.38$. The parameters of the dual exponential model for the glutamatergic AMPA and NMDA synapses, and the GABAergic synapses are given in Table 4.

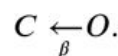
Finally, because the SPN model contains less electrical compartments than the number of synapses reported in Cheng et al. (1997), this implies that several background noise synapses will arrive in a single compartment. As such, their synaptic inputs will be integrated in the same voltage variable. Because of this, we have represented such multiple synapses arriving in one compartment with a single synapse instead, whose input frequency is scaled by the number of synapses it represents. This is why a non-saturating synapse model, such as the dual exponential model, was necessary for the background noise synapses. This also greatly increases simulation speed.

Saturating synapse model

The saturating synapse models are taken from Gao et al. (2021), which are a variation of the saturating synapse models in Destexhe et al. (1994). They are kinetic models which operate according to two different kinetic schemes depending on the presence or absence of neurotransmitters. When neurotransmitters arrive at the postsynaptic site, receptor dynamics are described by a kinetic scheme that models switching between closed (C) and open (O) states with rates of α and β :



where $[T]$ is the transmitter concentration. After neurotransmitter has been cleared from the synaptic cleft, receptor dynamics evolves according to a kinetic scheme that describes just the closing of the receptor with a rate β :



In this model, when a presynaptic spike arrives, neurotransmitter levels always reach a fixed saturating concentration in the synaptic cleft, $[T]_{\max}$, which lasts for a short time T_{dur} (a short glutamate lifetime in the synaptic cleft of cultured hippocampal synapses has been reported in Clements et al. (1992)). If another presynaptic spike arrives while the neurotransmitter pulse is still “on”, the pulse duration is lengthened by T_{dur} .

The synaptic conductance g_{syn} in this model is represented by the fraction of glutamate receptors in the open state, $[O]$:

$$g_{\text{syn}} = w g_{\max} [O],$$

where $g_{\max}$ is again the maximal synaptic conductance, and w is the synaptic weight which is modified by the learning rule. The parameters in the saturating synapse model are given in Table 5.

Table 4. The parameters in the dual exponential synaptic model.

Parameter	AMPA	NMDA	GABA
τ_A (ms)	1.9	2.76	0.1
τ_B (ms)	4.8	115.5	10
E_{rev} (mV)	0	15	-60
g_{max} (nS)	1.5	3.5	1.5

Table 5. The parameters in the saturating synapse model.

Parameter	AMPA	NMDA	eNMDA
E_{rev} (mV)	0	15	15
g_{max} (nS)	1.5	2.5	4.5
T_{dur} (ms)	1	1	50+200w
$[T]_{max}$ (mM)	1	1	0.02
α [(ms mM)]	12.5	4	4
β (/ms)	0.25	0.01	0.01
K_d (μ M)	20	2.5	2.5

Glutamate spillover

To produce plateau potentials which have an all-or-none jump in the voltage, i.e. a threshold non-linearity (as suggested by existing *in vitro* experiments), we included glutamate spillover. A detailed study of glutamate spillover and its role in triggering plateau potentials is given in Trpevski et al. (2023), and here we briefly describe the model. Glutamate spillover is a phenomenon that occurs during repeated synaptic stimulation, when glutamate reuptake by the surrounding astrocytes is saturated and when the astrocytes may even reverse function from reuptake to glutamate release (Malarkey and Parpura, 2008). When this happens in a synaptic cluster, glutamate spills over from the synaptic cleft into the extrasynaptic space, activating extrasynaptic NMDARs (eNMDARs). To model glutamate spillover, we place eNMDARs in the dendritic shaft, at the base of each spine in a cluster. Glutamate spillover is activated when a threshold level of clustered synaptic stimulation is reached – the “glutamate threshold”. In this study, the glutamate threshold is reached when 20 synapses with a weight $w > 0.5$ are activated (with 3 spikes each). Note that this threshold is equivalent to activating 10 synapses with a weight $w > 1.0$, or 40 synapses with a weight $w > 0.25$, i. e. the summed weight of all activated clustered synapses should be greater than 10.

When glutamate spills over, i) an activating spike to the eNMDARs is delivered 1.5 ms after the glutamate threshold is reached, ii) it stays much longer around the eNMDARs than in the synaptic cleft (T_{dur} is much longer), and iii) its concentration around the eNMDARs is much lower than in the synaptic cleft ($[T]_{\text{max}}$ is much lower) (Table 5 and Trpevski et al. (2023)). Also, the eNMDARs represent about 64% of the total NMDA conductance in this study (Table 5).

Stimulation protocol

The stimulation protocol in Fig. 3A for presenting the patterns (in random order), which are followed by dopamine feedback, contains several additional details that we list here. First of all, when the distributed or clustered synapses are activated within the 35 ms stimulation window, 3 input spikes arrive to each synapse within this window (as explained in the caption of Fig. 3). This is because at physiological extracellular calcium concentrations, single spikes do not induce plasticity (at least in the spike-timing-dependent protocols in Inglebert et al. (2020)). Furthermore, with the backward rate constants from Dorman et al. (2018) for calcium unbinding from the buffers are much lower than experimentally measured values for calmodulin and calbindin (Faas et al., 2011; Nägerl et al., 2000). To avoid long simulation times on the computing cluster used for this study, 600 ms after a pattern is presented, we reset the values of the buffers, their bound forms to calcium, and the intra- and extracellular calcium concentrations back to their initial values before the first pattern is presented, thus “equilibrating” the system faster and shortening simulation time. As can be seen from Fig. 2C, by 600 ms the plateaus are over, and both $[\text{Ca}]_{\text{NMDA}}$ and $[\text{Ca}]_{\text{L-type}}$ are back to their baseline levels, so this abrupt reset of the system does not affect learning and synaptic plasticity. And as can be seen from Fig. 3C, it neither affects the baseline voltage in the soma and dendrites before the arrival of a future pattern.

Performance scores on the FBP and NFBP

In the FBP, a score of 50% means that the SPN is correct only half of the time. This happens if the SPN is silent for both the relevant and irrelevant pattern (correct output only for the irrelevant pattern), or if it spikes for both patterns (correct output only for the relevant pattern). A score of 100% indicates it always spikes for the relevant pattern, and is always silent for the irrelevant pattern, and the FBP is perfectly solved. The threshold score for solving the FBP is 75%, exemplified with the situation where the SPN is always silent for the irrelevant pattern, and spikes at least half of the time for the relevant pattern. A score of 0% would indicate that the SPN's output is incorrect all the time, being silent for the relevant pattern, and spiking for the irrelevant pattern (this does not happen in the simulations).

Group 1

Up to 3 features per dendrite

Dendrite 1	Dendrite 2
	
	
	
	 
	 
	 
	 
	 
	 
	 
	
	
	
	 
	 
	 
	 
	 
	 

Group 2

4 features in at least 1 dendrite

Dendrite 1	Dendrite 2
	  
	  
	  
	  
	  
	  
	  
	  
	
	 
	 
	 
	 
	 
	 
	 
	 
	 
	 

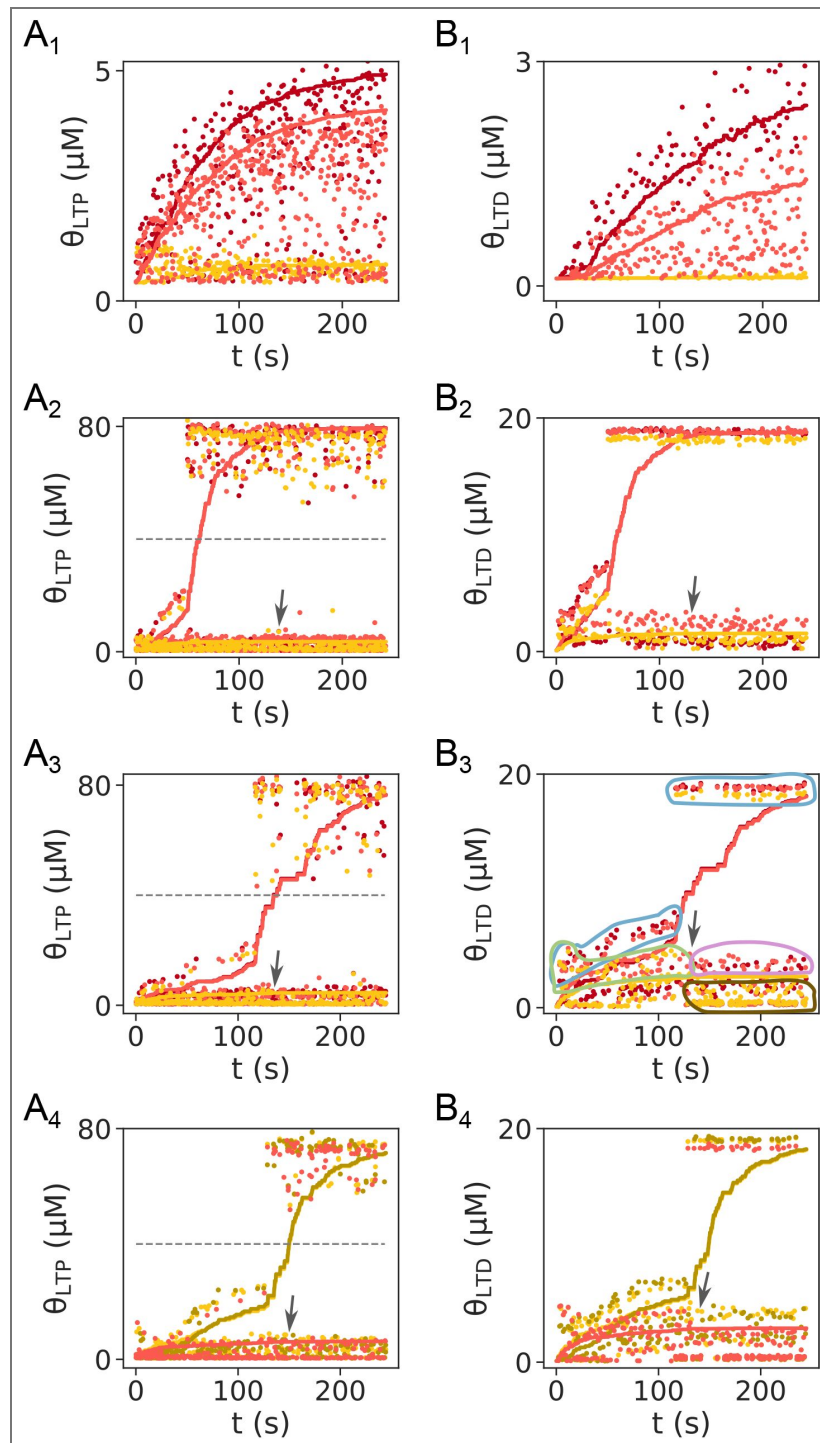


Figure 4—figure supplement 1. The calcium thresholds and calcium amplitudes from Fig. 4 shown in larger panels with greater detail.

(A) LTP thresholds for the FBP with distributed (A_1) and clustered (A_2) synapses, and for the NFBP (A_3 , A_4). (B) LTD thresholds for the FBP with distributed (B_1) and clustered (B_2) synapses, and for the NFBP (B_3 , B_4). In (B_3), regions with calcium amplitudes of interest are highlighted. (The same regions exist in all panels, although they are not highlighted.)

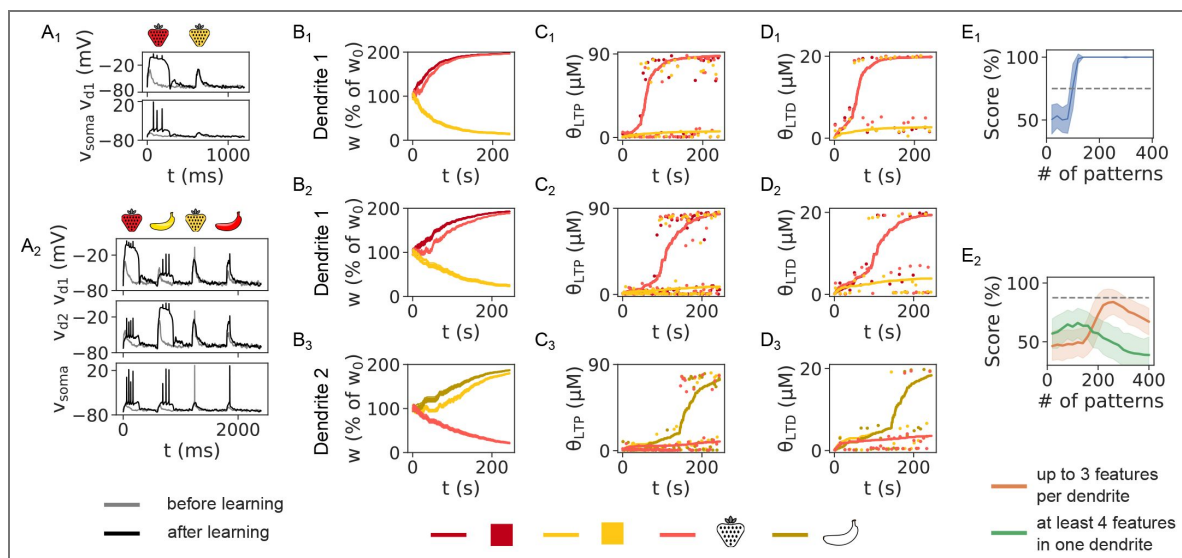


Figure 4—figure supplement 2. The effect of no upper threshold Θ_{LTP} on supralinear integration.

(A) The somatic and dendritic voltages before and after learning in the FBP (A_1) and NFBP (A_2). Not having an upper threshold causes somatic spiking for the irrelevant patterns after learning in the NFBP, but not the FBP. (B-D) The evolution of synaptic weights (B), LTP thresholds (C) and LTD thresholds (D) in the FBP (B_1 - D_1) and the NFBP (B_2 - D_2 and B_3 - D_3). Strengthened synapses saturate at their maximal levels, also driving more weakening in the weakened synapses. Having no upper threshold does not affect the thresholds of the strengthened synapses, but causes the thresholds of the weakened synapses to stabilize at higher levels. (E) Performance on the FBP (E_1) and NFBP (E_2).

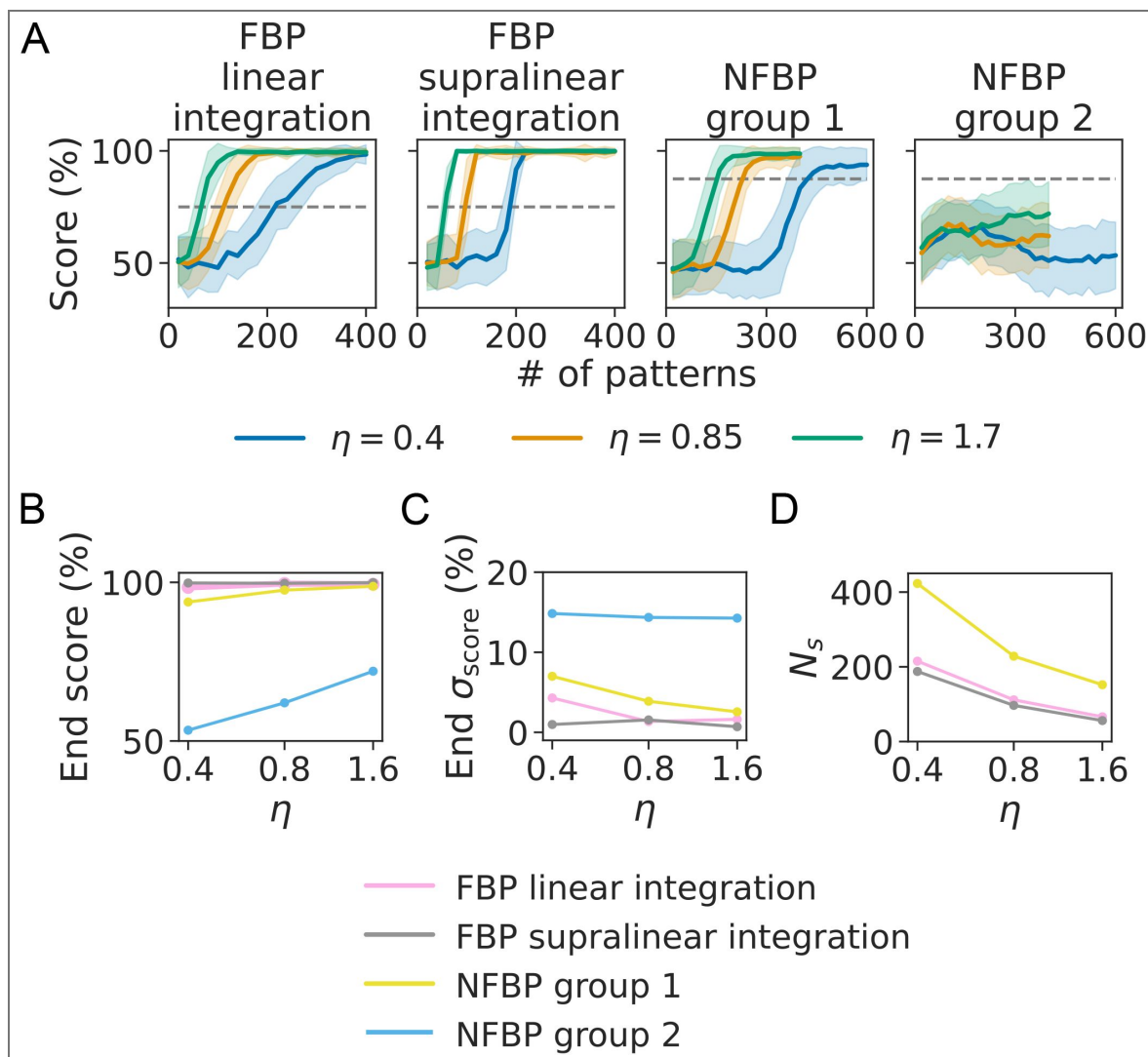


Figure 4—figure supplement 3. The effect of the learning rate η on learning.

(A) The average score on the FBP and NFBP for three values of the learning rate, $\eta \in \{0.4, 0.85, 1.7\}$. Increasing the learning rate causes faster learning of the tasks (less pattern presentations are needed). (B) The score at the end of learning varies little with η , except in group 2 for the NFBP, where a higher learning rate increases performance, so that one of the relevant patterns is remembered. (C) The standard deviation at the end of learning also varies little with η (there is a 5% decrease for the NFBP, group 1 only). (D) The speed of learning measured by the number of patterns, N_s , needed to reach the score threshold for solving each task (the dashed line in (A)). The speed of learning to solve the tasks is affected by the learning rate η , as is also seen in (A). Increasing the learning rate speeds up learning.

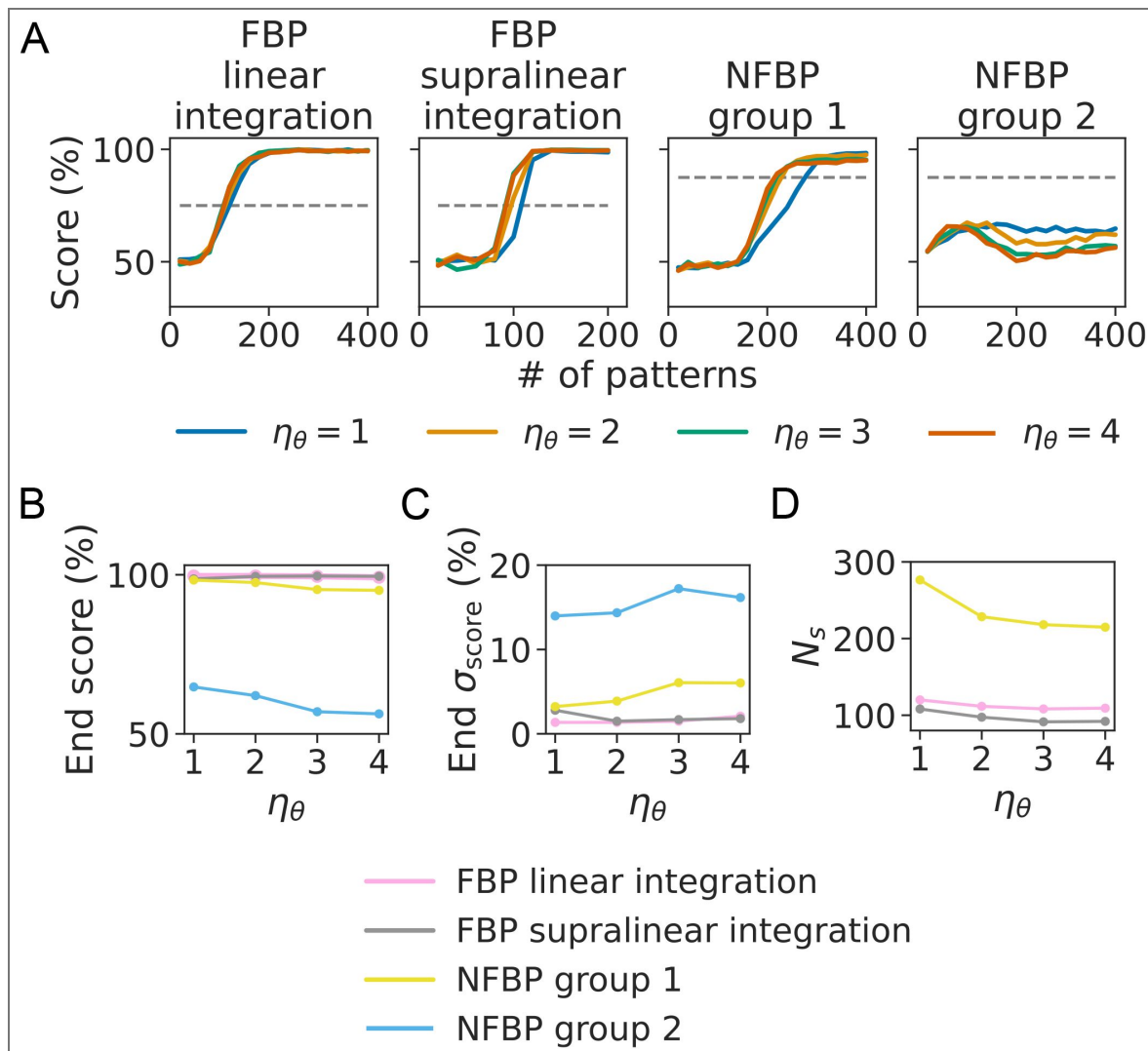


Figure 4—figure supplement 4. The effect of the metaplasticity rate η_{θ} on learning.

(A) The average score on the FBP and NFBP for four values of the metaplasticity rate, $\eta_{\theta} \in \{1, 2, 3, 4\}$ (the standard deviation is omitted for figure clarity). There is little effect of the metaplasticity rate on the final score (B), the final standard deviation of the score (C) and the speed of learning (D). (B) The score at the end of learning varies little with η_{θ} (only slightly for the NFBP). (C) The standard deviation at the end of learning also varies little with η_{θ} (up to 5% increase for the NFBP only). (D) The speed of learning measured by the number of patterns, N_s , needed to reach the score threshold for solving each task (the dashed line in (A)). The speed of learning to solve the FBP varies very little with η_{θ} , while that for solving the NFBP only shows a large increase between $\eta_{\theta} = 1$ and $\eta_{\theta} = 2$.

Figure 5—figure supplement 1. Learning outcome on the FBP with linear (A_1 – E_1) and supralinear integration (A_2 – E_2) without metaplasticity in the LTD threshold.

(A) The somatic and dendritic voltage evoked by both patterns before and after learning. (B, C) The evolution of synaptic weights (B) and LTP thresholds (C) during learning. (D) The LTD thresholds are fixed. In (B) all synapses are shown, and in (C, D) only one synapse per feature is used to show its calcium threshold. (E) The performance on the FBP.

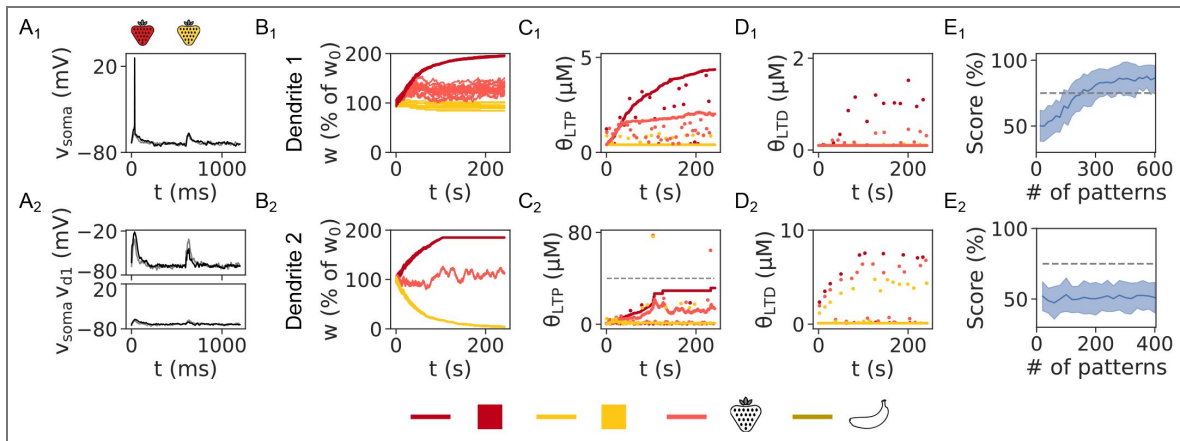


Figure 6—figure supplement 1. Learning outcome on the FBP with linear (A_1 – E_1) and supralinear integration (A_2 – E_2) without metaplasticity in the LTP threshold.

(A) The somatic and dendritic voltage evoked by both patterns before and after learning. (B) The evolution of synaptic weights during learning. (C) The LTP thresholds are fixed. (D) The evolution of the LTD thresholds during learning. (E) The performance on the FBP.

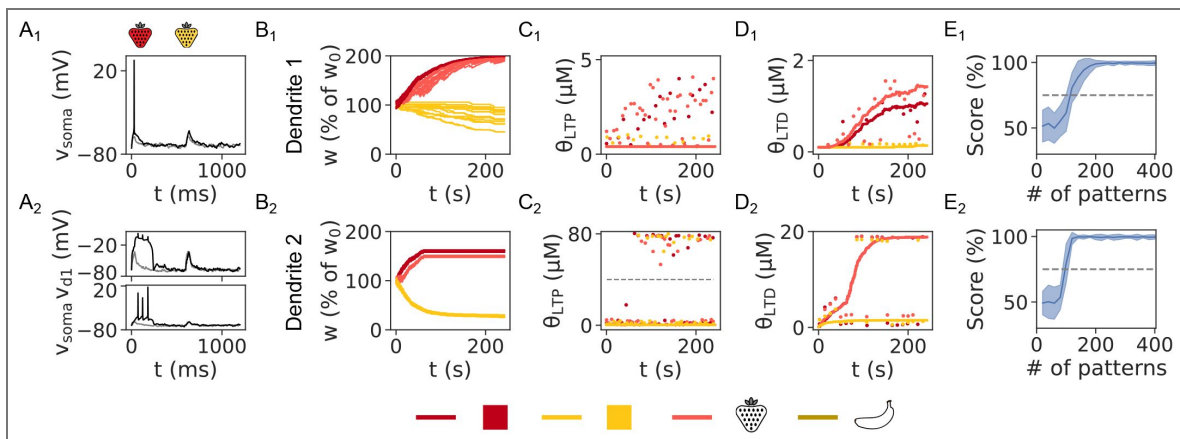


Figure 7—figure supplement 1. Learning outcome on the FBP with linear (A₁-E₁) and supralinear integration (A₂-E₂) with partial metaplasticity.

(A) The somatic and dendritic voltage evoked by both patterns before and after learning. (B-D) The evolution of synaptic weights (B), LTP thresholds (C) and LTD thresholds (D) during learning. In (B) all synaptic weights are shown, and in (C, D) the thresholds for only one synapse per feature are shown. (E) The performance on the FBP.

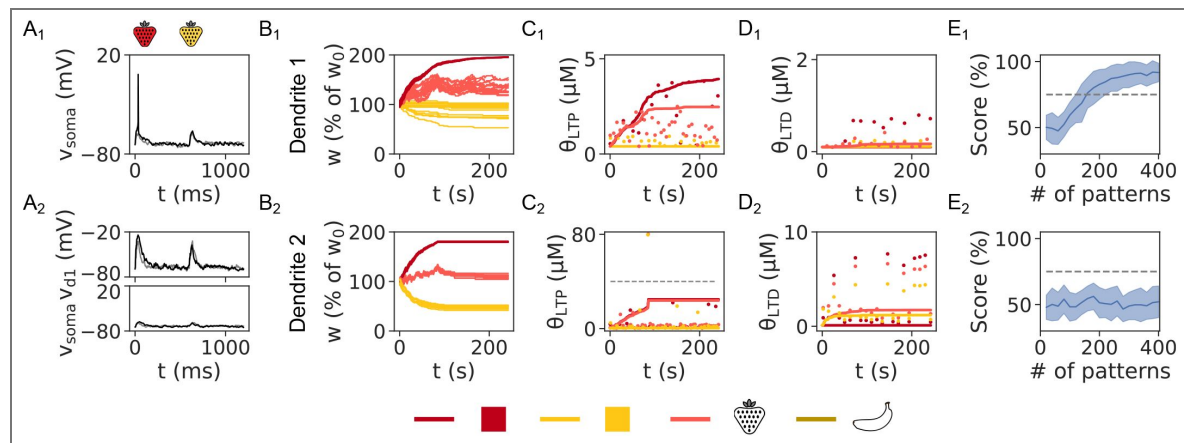
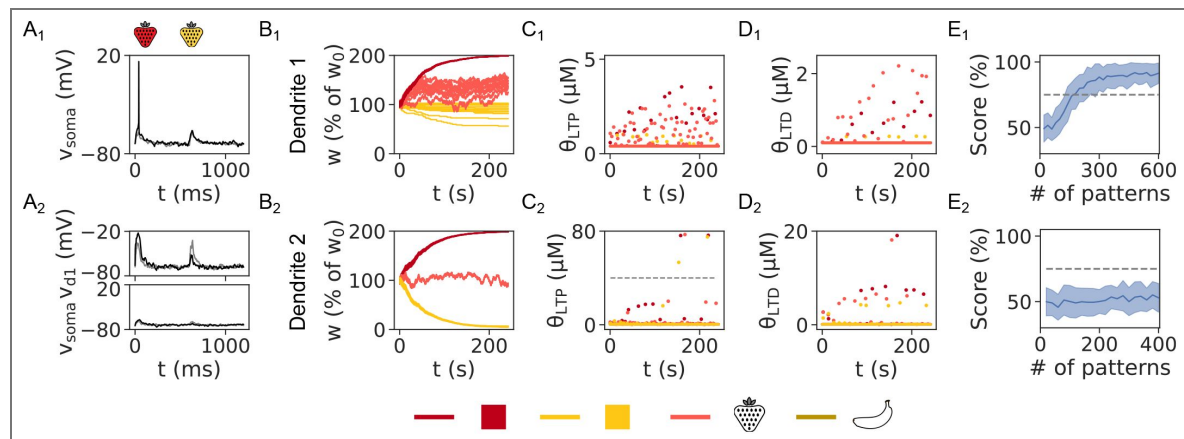


Figure 8—figure supplement 1. Learning outcome on the FBP with linear (A₁-E₁) and supralinear integration (A₂-E₂) without metaplasticity.

(A) The somatic and dendritic voltage evoked by both patterns before and after learning. (B) The evolution of synaptic weights during learning. (C, D) The LTP (C) and LTD thresholds (D) are fixed. In (B) all synaptic weights are shown, and in (C, D) the thresholds for only one synapse per feature are shown. (E) The performance on the FBP.



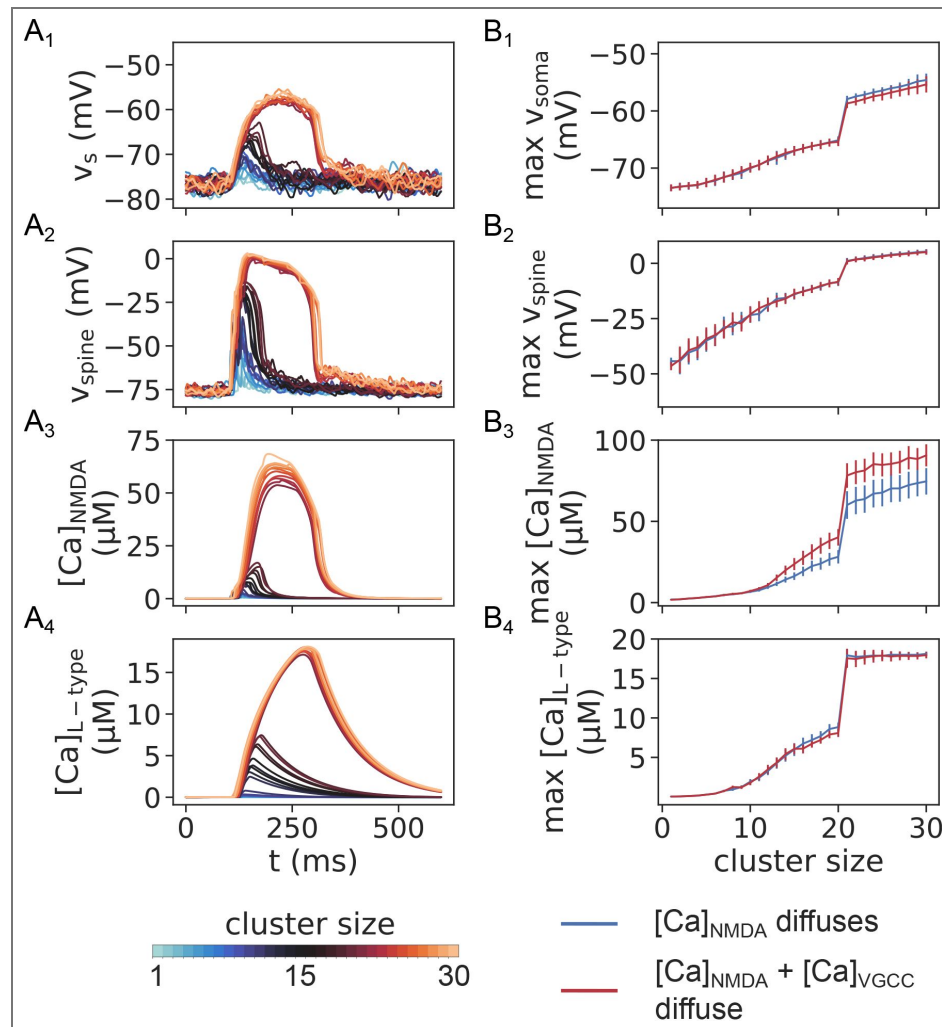


Figure 9—figure supplement 1. Voltage and calcium elevations when only $[Ca]_{NMDA}$ diffuses axially through the SPN.

(A₁-C₄) Somatic voltage, spine voltage, spine $[Ca]_{NMDA}$ and spine $[Ca]_{L-type}$ evoked in the supralinear integration scenario, to be compared with Fig. 2C (synaptic cluster placed on one dendrite, in a 20-micrometer region approximately 120 micrometers away from the soma). Line color represents the size of the synaptic cluster, which is varied from 1 to 30 synapses. (B₁-B₄) The amplitude of the somatic voltage, spine voltage, spine $[Ca]_{NMDA}$ and $[Ca]_{L-type}$ compared between the cases of only $[Ca]_{NMDA}$ diffusing and $[Ca]_{NMDA} + [Ca]_{VGCC}$ diffusing (the latter is replfrom Fig. 2D). Results are averages over 20 trials, and in the case of synaptic clusters, over 8 different dendrites (clusters located in a 20-micrometer region starting at approximately between 120 micrometers from the soma).

Data availability

The current manuscript is a computational study, so no data have been generated for this manuscript. Modelling code is openly accessible at <https://github.com/danieltrpevski/Plasticity/tree/metaplasticity>

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Peer reviews

Reviewer #1 (Public review):

Summary:

This computational modelling study addresses the important question of how neurons can learn non-linear functions using biologically realistic plasticity mechanisms. The study extends the previous related work on metaplasticity by Khodadadi et al. (2025), using the same detailed biophysical model and basic study design, while significantly simplifying the synaptic plasticity rule by removing non-linearities, reducing the number of free parameters, and limiting plasticity to only excitatory synapses. The rule itself is supervised by the presence or absence of a binary dopamine reward signal, and gated by separate calcium-

sensitive thresholds for potentiation and depression. The author shows that, when paired with a strong form of dendritic non-linearity called a "plateau potential" and appropriate pre-existing dendritic clustering of features, this simpler learning mechanism can solve a non-linear classification task similar to the classic XOR logic operator, with equal or better performance than the previous publication. The primary claims of this publication are that metaplasticity is required for learning non-linear feature classification, and that simultaneous dynamics in two separate thresholds (for potentiation and depression) are critical in this process. By systematically studying the properties of a biophysically plausible supervised learning rule, this paper adds interesting insights into the mechanics of learning complex computations in single neurons.

Strengths:

The simplified form of the learning rule makes it easier to understand and study than previous metaplasticity rules, and makes the conclusions more generalizable, while preserving biological realism. Since similar biophysical mechanisms and dynamics exist in many different cell types across the whole brain, the proposed rule could easily be integrated into a wide range of computational models specializing in brain regions beyond the striatum (which is the focus of this study), making it of broad interest to computational neuroscientists. The general approach of systematically fixing or modifying each variable while observing the effects and interactions with other variables is sound and brings great clarity to understanding the dynamic properties and mechanics of the proposed learning rule.

Weaknesses:

General notes

- (1) The credibility of the main claims is mainly limited by the very narrow range of model parameters that was explored, including several seemingly arbitrary choices that were not adequately justified or explored.
- (2) The choice to use a morphologically detailed biophysical model, rather than a simpler multi-compartment model, adds a great deal of complexity that further increases uncertainty as to whether the conclusions can generalize beyond the specific choices of model and morphology studied in this paper.
- (3) The requirement for pre-existing synaptic clustering, while not implausible, greatly limits the flexibility of this rule to solve non-linear problems more generally.
- (4) In order to claim that two thresholds are truly necessary, the author would have to show that other well-known rules with a single threshold (e.g., BCM) cannot solve this problem. No such direct head-to-head comparisons are made, raising the question of whether the same task could be achieved without having two separate plasticity thresholds.

Specific notes

(1) Regarding the limited hyperparameter search:

(a) On page 5, the author introduces the upper LTP threshold Θ_{LTP} . It is not clear why this upper threshold is necessary when the weights are already bounded by w_{max} . Since w_{max} is just another hyperparameter, why not set it to a lower value if the goal is to avoid excessively strong synapses? The values of w_{max} and Θ_{LTP} appear to have been chosen arbitrarily, but this question could be resolved by doing a proper hyperparameter search over w_{max} in the absence of an upper Θ_{LTP} .

(b) The author does not explore the effect of having separate learning rates for Θ_{LTP} and Θ_{LTD} , which could also improve learning performance in the NFBP. A more

comprehensive exploration of these parameters would make the inclusion of $\theta_{\max}$ (and the specific value chosen) a lot less arbitrary.

(c) Figure 4 Supplements 3-4: The author shows results for a hyperparameter search of the learning rule parameters, which is important to see. However, the parameter search is very limited: only 3 parameter values were tried, and there is no explanation or rationale for choosing these specific parameters. In particular, the metaplasticity learning rates do not even span one order of magnitude. If the author wants to claim that the learning rule is insensitive to this parameter, it should be explored over a much broader range of values (e.g., something like the range [0.1-10]).

(2) Regarding the similarity to BCM, the author would ideally directly implement the BCM learning rule in their model, but at the least the author could have shown whether a slight variant of their rule presented here can be effective: for example having a single (plastic, not fixed) Ca-dependent threshold that applies to both LTP and LTD, with a single learning rate parameter.

(3) This paper is extremely similar (and essentially an extension) to the work of Khodadadi et al. (2025). Yet this paper is not mentioned at all in the introduction, and the relation between these papers is not made clear until the discussion, leaving me initially puzzled as to what problems this paper addresses that have not already been extensively solved. The introduction could be reworked to make this connection clearer while pointing out the main differences in approach (e.g., the important distinction between "boosting" nonlinearities and plateau potentials).

(4) The introduction is missing some citations of other recent work that has addressed single-neuron non-linear computation and learning, such as Gidon et al (2020); Jones & Kording (2021).

(5) Figure 1: The figure prominently features mGluR next to the CaV channel, but there is no mention of mGluR in the introduction. The introduction should be updated to include this.

(6) Could the author explain why there is a non-monotonic increase/decrease in the $[Ca]_L$ in Figure 2B_4? Perhaps my confusion comes from not understanding what a single line represents. Does each line represent the $[Ca]$ in a single spine (and if so, which spine), or is each line an average of all the spines in a given stim condition?

(7) Row 124 (page 4): L-type Ca microdomains (in which ions don't diffuse and therefore don't interact with Ca_NMDA) is a critical assumption of this model. The references for this appear only in the discussion, so when reading this paper, I found myself a bit confused about why the same ion is treated as two completely independent variables with separate dynamics. Highlighting the assumption (with citations) a bit more clearly in the results section when describing the rule would help with understanding.

(8) Row 149 (page 5): The current formulation of the update rule is not actually multiplicative. The fact that the update is weight-dependent alone does not make it a multiplicative rule, and judging by equation (1) it appears to simply be an additive rule with a weight regularization term that guarantees weight bounds. For example, a similar weight-dependent update is also a core component of BTSP (Milstein et al. 2021; Galloni et al. 2025), which is another well-known *additive* rule. An actual multiplicative rule implies that the update itself is applied via a multiplication, i.e. $w_{\text{new}} = w_{\text{old}} * \Delta w$

For an example of a genuinely multiplicative rule, see: Cornford et al. 2024, "Brain-like learning with exponentiated gradients"). Multiplicative rules have very different properties to additive rules, since larger weights tend to grow quickly while small weights shrink towards 0.

(9) Equation 1 (page 5): Shouldn't the depression term be written as: $(w_{\min} - w)$? This term would be negative if w is larger than $w_{\min}$, leading to LTD. As it is written now, a large w and small $w_{\min}$ would just cause further potentiation instead of depression.

(10) In the introduction, the teaching signal is described in binary terms (DA peak, or DA pause), but in Equation 1, it actually appears to take on 3 different values. Could the author clarify what the difference is between a "DA pause" and the "no DA" condition? The way I read it, pause = absence of DA = no DA

(11) Figure 3: In these experimental simulations, DA feedback comes in 400ms after the stimulus. The author could motivate this choice a bit better and explain the significance of this delay. Clearly, the equations have a δ_t term, but as far as the learning algorithm is concerned, it seems like learning would be more effective at $\delta_t=0$. Is the choice of 400ms mainly motivated by experimental observations? On a related note, is it meaningful that the 200ms δ_t before the next stimulus is shorter than the 400ms pause from the first stimulus? Wouldn't the DA that arrives shortly before a stimulus also have an effect on the learning rule?

(12) Figure 4C: How is it possible that the θ_{LTP} value goes higher than the upper threshold (dashed line)? Equation 3 implies that it should always be lower.

(13) Row 429 (page 11): The statement that "without metaplasticity the NFBP cannot be solved" is overly general and not supported by the evidence presented. There exist many papers in which people solve similar non-linear feature learning problems with Hebbian or other bio-plausible rules that don't have metaplasticity. A more accurate statement that can be made here is that the specific rule presented in this paper requires metaplasticity.

(14) The methods section does not make any mention of publicly available code or a GitHub repository. The author should add a link to the code and put some effort into improving the documentation so that others can more easily assess the code and reproduce the simulations.

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Reviewer #2 (Public review):

Summary:

The manuscript proposes interesting synaptic plasticity rules grounded in experimental data. Its main features are:

(1) plasticity depends on local calcium concentration driven by presynaptic activity and is independent of somatic action potentials,

(2) the rules incorporate metaplasticity, and (3) they demonstrate how a single neuron could address the feature-binding problem at the dendritic level.

The work extends a previous study (<https://doi.org/10.7554/eLife.97274.2> ) , to which the author also contributed.

The author models two calcium thresholds (LTP/LTD) from two different calcium sources (NMDA/VGCC), and these thresholds are flexible (metaplasticity rule, similar to BCM), which is claimed to be necessary for successful learning of both FBP and NFBP (linear and nonlinear feature binding problem with 1 or 2 patterns). The role of each threshold seems to be opposite and complementary. One extra condition has been added: an upper threshold for LTP. This threshold serves to stop synaptic strengthening once synapses are strong enough to evoke a plateau. With that, synapses are not strengthened to the maximal value, avoiding strong supralinear integration for irrelevant patterns.

Strengths:

The current model implements not only local synaptic plasticity but also metaplasticity and solves the FBP at the dendrite level. Another strong aspect of the model is that metaplasticity in the LTD threshold protects strengthened synapses from weakening. In this way, as the author mentioned, metaplasticity is able to protect learned patterns from being forgotten or weakened and prevent irrelevant patterns from being stored. This is a nice modelling example of metaplasticity being helpful in preventing the catastrophic interference or forgetting (as has been explicitly discussed in a recent article <https://doi.org/10.1016/j.tins.2022.06.002>). The author might want to briefly mention or emphasize this aspect of the model, which might be interesting also for the AI community.

Weaknesses:

(1) What is novel in the current paper as compared to Khodadadi et al. eLife 2025? That is not completely clear and should be made clearer. Is it only a minor difference related to the fact that the new learning rule has metaplasticity in both calcium thresholds and is simpler? This seems to be just an incremental increase in knowledge/methods. Can the author defend his paper against this point from the „devil's advocate"? How is the conclusion of the author in the abstract that „metaplasticity in both thresholds is necessary" reconcilable with his previous publication (Khodadadi et al. eLife 2025), in which only metaplasticity in one threshold was successful in solving the nonlinear feature binding problem?

(2) As far as I can judge without testing the model, metaplasticity causes thresholds to monotonically increase during systematic pattern presentation, which stabilizes weights and allows pattern separation. Due to the closed-loop nature of the current implementation, where metaplasticity only happens if plasticity happens, this also effectively locks patterns in place. However, flexible learning is an essential mechanism for survival. Imagine a mutation event takes place and bananas suddenly become red and/or strawberries turn yellow. It seems that the current model would be unable to adapt to these new patterns even if rewards were to be shifted. While out of the scope of the study, due to its importance, I feel that pattern shifting/relearning should at least be briefly discussed. How could the model be improved to allow relearning?

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